Interoceptomics Technical Script

The slide ranges follow the 111 presentation pages and their 135 reveals. Each section develops the mechanism behind the visible words. The experiments describe a research programme; proposed interfaces and results remain proposals unless the cited evidence establishes them.

I Know Kung Fu

Slide 1 | Reveals 1

I know kung fu. Four words, and years of practice disappear. In The Matrix, information enters through a cable and Neo emerges with a skill. The film skips the part that matters to us: what changed in the nervous system, and how would anybody check?

A connection could supply instructions. It could help control the current movement. It might eventually help a learner acquire an ability that remains after the instructions stop. Those are three different outcomes. A device that lets someone act while it is connected can be valuable in its own right; calling that result an uploaded skill would add something the test had never measured.

So I want to begin smaller than the film. Choose a movement whose position, timing and success can be measured independently. Record specified signals during that movement. Ask whether a model predicts the outcome on trials it has never seen. If a later feedback system helps a recipient, compare the recipient's performance during assistance with performance after trainer-specific guidance ends. Necessary compensatory support would stay unchanged, so that removing help does not simply remove the ability to act at all.

Software fitting and biological learning have different subjects. Falling prediction error means that software improved on a stated test. Lasting improvement in a person or animal needs its own evidence. The distance between those claims is where the experiment begins.

Reveal Cues

1. Show the title, image and subtitle together. Explain Neo and the BCI subtitle together; distinguish instructions, assistance and retained skill.

References

The Matrix (1999), official film page https://www.warnerbros.com/movies/matrix

Brain2Body programme ethos, current source

High-Fidelity Telepathy

Slide 2 | Reveals 2

How would you get information from one nervous system into another? Professor X supplies a wonderfully economical answer: think harder. An experiment needs a route, a signal and an observable consequence.

Here, high fidelity would mean fidelity to a specified target. Perhaps a channel helps a recipient distinguish one task cue from another; perhaps it improves a defined movement estimate. We would need the information available to the sender, the observation actually recorded, the transformation applied, the delivery method and the recipient's response. A good result at one link cannot certify the remaining links. Recording an accurate signal says nothing yet about whether a recipient can use it.

Bandwidth is only part of that problem. More samples can reproduce the wrong feature more faithfully. Timing matters too: a useful instruction arriving after the action is over is faithfully delivered and functionally late. Compare aligned information with an appropriately shuffled or delayed version, alongside no-link and sham conditions where justified, and measure the task independently of the channel's own score.

Telepathy remains the provocative question, not the result being announced. Even successful transfer of task information would leave identical private experience, arbitrary thought reading and lasting skill transfer unresolved. The useful claim is the one the recipient's behavior can support. Before building an artificial route, though, there is a remarkable natural connection worth examining.

Reveal Cues

1. Show the title, image and subtitle together. Use Professor X to introduce the question; define task-specific fidelity and an observable response.

References

Brain2Body programme ethos, current source

It's Already Here

Slide 3 | Reveals 3

A hand draws another hand. The drawing is on paper; the feedback is in the act. Each movement changes the mark we see, and what we see can change the next movement. Nervous systems make that return path possible: action changes sensation, and sensation changes the next action.

That does not make every feedback loop telepathy. It gives us a way to ask a much more concrete question. If information reaches a nervous system through an unusual connection, can we identify what crossed and what the recipient did with it? We can measure responses without pretending that a response reveals a complete private experience.

Think of two people following the same pointing gesture. Both can turn toward the correct object while noticing different things about it. Agreement about the object is evidence of useful communication; it does not establish identical perception. A laboratory link faces the same distinction, with additional uncertainty about which features of the sender's activity were recorded and how the receiver interpreted the delivered information.

The attraction of a closed loop is that an answer has consequences. The recipient acts, the world changes, and the resulting observation can challenge the prediction. An offline recording offers a different, narrower test: whether a model predicts a reserved observation. Both can teach us something. Neither should quietly inherit the other's achievements. The natural example that follows motivates the question; it does not finish the engineering.

Reveal Cues

1. Show the title, image and subtitle together. Explain the hand-drawing image and subtitle; explain action, sensation and the limits of response agreement.

References

Brain2Body programme ethos, current source

How They Think

Slide 4 | Reveals 4

Krista and Tatiana Hogan are two people, with two brains and an extraordinary connection between them. Each has her own life and perspective. Their case asks what information a physical connection can make available to someone else, and how we could tell.

The documentary Inseparable reports experiences involving sight, taste and touch across their connection. Those accounts are the reason the case belongs here. They suggest that information originating on one side can sometimes be available to the other. They do not supply a calibrated channel capacity, a complete map of the connection or a controlled demonstration of arbitrary thought transfer.

We should ask what each observation actually distinguishes. Did one person identify a stimulus she could not otherwise access? Could shared movement, sound, expectation or familiar routines contribute? What timing and comparison would separate those explanations? These questions do not dismiss the reports. They tell us what additional evidence would turn a compelling case into a specific functional model.

Nor are the twins a ready-made design for an implant. Their connection developed as part of their bodies and lives. An artificial interface would have different sampling, delivery and adaptation limits. The lesson is a question we can carry into the lab: can a specified route convey useful information between distinct nervous systems? Their reported experience motivates that question without making the proposed apparatus equivalent to them.

Reveal Cues

1. Show the title, image and subtitle together. Explain the twins photo and exact subtitle; name Krista and Tatiana and identify documentary-report limits.

References

Curious Features, Inseparable documentary https://curiousfeatures.ca/project/inseparable/

Share A Brain

Slide 5 | Reveals 5

Share a brain is the headline's shorthand. Documentary accounts describe a connection between the twins' thalamic regions while each has her own brain. The 2010 Channel 4 programme follows the family's account and the clinicians investigating it; the Wellcome catalogue names Doug Cochrane among its contributors. That establishes the documentary context, not an independently inspected clinical scan. The CBC documentary illustration depicts their connected anatomy; it is not a clinical scan, and its precise patient-specific anatomy remains unverified here.

A physical connection makes sensory sharing a tractable possibility. Yet tissue continuity alone does not tell us which signals travel, their direction, their timing or their interpretation. A bridge can carry information selectively. The recipient also has a developing nervous system which has learned to live with that connection. The same input delivered to a different system might have a different consequence.

Sight, taste and touch therefore stay attached to reported observations. To study a corresponding engineered channel, we would choose a defined stimulus and an independently scored response. We would record what the sender could observe and what the receiver could access through every other route. Aligned, shuffled and no-link comparisons could then test whether the channel added information beyond those alternatives.

Even a successful result would describe a relationship among stimulus, route and response. It would not show that red looks identical to both people, that a taste has been copied intact, or that a memory crossed with it. The ambition becomes useful when we stop asking a bridge to prove every claim at once. First find out what, if anything, the connection lets the recipient do.

Reveal Cues

1. Show the title, image and subtitle together. Explain the thalamic-bridge illustration and sensory subtitle; retain reported/schematic scope.

References

Curious Features, Inseparable documentary https://curiousfeatures.ca/project/inseparable/

Brain2Body programme ethos, current source

Wellcome Collection, The twins who share a brain, Channel 4 Body Shock, broadcast 25 May 2010 https://wellcomecollection.org/works/qugqdhqy

Want To Recreate This

Slide 6 | Reveals 6

I'd like to recreate a bounded version of that information link. There is a problem lurking underneath the appealing picture: we do not yet know the code we would need to reproduce.

A nervous recording mixes the event we care about with body state, task history, measurement noise and activity we have not explained. A learned mapping can exploit any of those. It may predict which turn comes next because every trial follows the same schedule. It may recognize the recording session instead of the movement. Give it new conditions and the apparent understanding can disappear.

We therefore need the recording and the event it is supposed to explain. The clock must align them, the sensors must have known limits, and the test must hold out the unit across which we claim generalization. Holding out frames next to training frames is a weak test of a new animal or a new task. An independent measurement of movement gives the prediction somewhere to be wrong; a simpler baseline tells us whether the extra machinery earned its place.

This is the hidden crocodile: a description can be impressive while the functional explanation remains missing. Biology has already met that difficulty at another scale. Reading the letters of DNA was an enormous achievement. Turning those letters into a dependable account of what a living system does has proved a longer journey.

Reveal Cues

1. Show the title, image and subtitle together. Explain the lurking crocodile and subtitle; expose shortcut learning and the need for independent targets.

References

Brain2Body programme ethos, current source

Human Genome Project Failed

Slide 7 | Reveals 7

The Human Genome Project failed. Against its sequencing objective, that is wrong. The project delivered the first human reference sequence and completed its programme in 2003. That achievement changed biology and medicine. The unfinished promise is what comes after the map.

The subtitle compresses two different statistics too far. The FDA's October 2026 framework estimates 300 million people worldwide living with rare diseases, and says approximately 5 percent of recognized rare diseases had an FDA-approved treatment as of 2025. People are one denominator; conditions are another. Those numbers do not establish that 300 million people receive no treatment, no useful care or no benefit from genomic medicine. Nor are all rare conditions genetic. The World Health Organization also estimates that more than 300 million people live with rare diseases. An NIH fact sheet discusses the share of rare diseases without FDA-approved treatments; neither figure establishes the share of people without effective or supportive care.

The defensible criticism is the gap between describing a system and intervening effectively in it. A sequence constrains possible mechanisms. Regulation, cell state, development and environment help determine the consequences of a variant, and a promising mechanism still needs evidence that changing it improves an outcome. Functional genomics has spent decades working on precisely those relationships; the field did not stop at a list of letters.

The achievement and the unmet need belong in the same account. We finished a foundational measurement. We have not finished the medicine. Neuroscience should learn from that distinction before treating its own maps as finished explanations. More complete description can accelerate the next experiment without completing the functional story.

Reveal Cues

1. Show the title, image and subtitle together. Explain DNA image and subtitle; explain the 300 million people versus~5% conditions denominators.

References

NHGRI, The Human Genome Project https://www.genome.gov/human-genome-project

FDA, A New Framework for Rare Disease Review, October 2026 https://www.fda.gov/media/195694/download?attachment=

WHO, World Health Assembly daily update, 24 May 2025 https://www.who.int/news/item/24-05-2025-seventy-eighth-world-health-assembly---daily-update--24-may-2025/

NIH NCATS, Rare Diseases Fact Sheet https://ncats.nih.gov/sites/default/files/NCATS_RareDiseasesFactSheet.pdf

2.5 Decades

Slide 8 | Reveals 8

Two and a half decades frames the period around and after the sequencing project. The 2003 completion to 2026 is about 23 years; the slide is a rounded interval, not a universal waiting period for biology. Knock-in and knockout experiments also have histories beyond this interval, and no single timetable describes every gene or disease.

A knockout inactivates or removes a specified gene. A knock-in introduces a defined sequence change. Comparing a changed system with an appropriate control can reveal contributions that observation alone misses. The word contribution matters: disrupting one component can alter development, trigger compensation or affect several processes. A changed outcome does not automatically identify one exclusive job for that gene.

Imagine removing one participant from a relay team and timing the race. The result tells you something about that team under that intervention. It cannot, by itself, tell you how the same participant would perform in another team, how the remaining runners adapted, or which part of the exchange caused the delay. The analogy explains the inferential problem; it supplies no biological result.

The lesson carries into neural experiments. We need comparisons that expose mechanisms, yet every intervention has a context and possible side effects. Repeating a narrow test can sharpen its answer. It cannot silently expand the answer to every condition. Speed matters, but the useful acceleration is more informative experiments per unit effort, with their original limits intact.

Reveal Cues

1. Show the title, image and subtitle together. Explain the rounded interval, controlled genetic perturbation and context.

References

NHGRI, Knockout https://www.genome.gov/genetics-glossary/Knockout

Brain2Body programme ethos, current source

Incremental Improvement

1 Gene At A Time

Slide 9 | Reveals 9

DNA made of Lego is a tempting picture. Swap one brick and see what the building does. For a specified question, that intervention can be informative. Living systems make the picture less tidy because the effect of a change depends on what else is present and when the change acts.

One gene at a time therefore describes an experimental approach, not the full scope of modern genetics. Contemporary research also studies many changes in parallel, combinations of changes and multiple molecular readouts. A fair argument cannot erase that work merely to make our own proposal sound new. The remaining difficulty is connecting those observations into a model which predicts a defined consequence under new conditions.

Suppose two components each appear dispensable when changed separately, while changing both disrupts the outcome. The separate tests did not discover that interaction. Conversely, a large collection of associations may leave causality unresolved. Breadth of measurement and strength of explanation answer different questions; the best design makes them work together.

For Brain2Body the equivalent is a growing set of signal channels. Adding more channels helps only if they improve a prespecified prediction beyond a simpler comparison, with uncertainty and genuinely reserved observations. Thousands of inputs can still conceal the missing variable. Incremental work becomes cumulative when its observations can be aligned, its assumptions inspected and its predictions tested against the same biological question.

Reveal Cues

1. Show Incremental Improvement, the image and 1 Gene At A Time together. Explain the one-gene example, interactions and modern multiplexed work.

References

NHGRI, Gene Regulation https://www.genome.gov/genetics-glossary/Gene-Regulation

Brain2Body programme ethos, current source

Genes ≠ Traits

Slide 10 | Reveals 10

A gene is not a trait. An observable trait can depend on genomic variation, regulation, development and environment; the relationship differs from one trait or condition to another. Some variants have large, well-established effects. Others change risk only in combination with other factors. The point is to identify the relationship, not to deny genetics.

A neural connection map presents a related challenge. Structure constrains what can happen, but activity, timing, inputs and body state help determine what actually happens in a task. Measuring a map and testing a functional prediction are complementary experiments. A more complete map may make a better test possible; it does not supply the functional result by itself.

The mistake would be to assume a one-to-one translation before checking it. State the observed trait or action, the information supplied to the model and the conditions on which it is tested. Then report what the model predicts, where it fails and whether competing explanations remain.

Reveal Cues

1. Show the title, image and subtitle together. Explain the gene–trait distinction and connect the analogy to an independently tested neural prediction.

References

NHGRI, Phenotype https://www.genome.gov/genetics-glossary/Phenotype

Brain2Body programme ethos, current source

Form ≠ Function

Slide 11 | Reveals 11

A crocodile and a Croc can share a name without sharing a job. A shoe's outline can suggest something about its use; you still have to put weight on it. Form constrains function, but inspecting the outline is not the same experiment as testing what it does.

The same caution applies to a connection map. It tells us which elements appear connected within a stated coverage and resolution. Predicting activity also requires a description of the inputs, the changing state and the dynamics of those connections. Several dynamical models can fit the same arrangement and disagree about the next event. Additional observations are what let us decide among them.

Function equals function is an insistence on a functional test, rather than a claim that anatomy is irrelevant. Ask a circuit to predict a specified response. Compare its prediction with an independent recording under declared conditions. When the question is causal, use appropriately justified interventions and competing explanations; correlation in a recorded task is insufficient on its own.

A model can fail usefully. The failure can tell us whether we need better structural coverage, an omitted state variable, a different body model or a correction to the observation itself. That is a more productive relationship between form and function: structure supplies constraints, measured consequences test the account. Neither gets to substitute for the other merely because one is easier to display.

Reveal Cues

1. Show the title, image and subtitle together. Explain Crocs and Function=Function; explain a functional test and the continuing value of anatomy.

References

Brain2Body programme ethos, current source

Disorder To Disease

Slide 12 | Reveals 12

The image shows virtual Atlas humanoids training in parallel in simulation. Many simulated trials can help develop a control model; they do not establish biological accuracy or clinical benefit. Look to AI because the data are large and their relationships are difficult, not because the label solves disease. A model can help represent variation, combine observations and predict outcomes. Whether that helps a patient is a further empirical question.

Disorder to disease asks us to connect a useful description of disrupted function with a clinically meaningful condition. The mapping cannot be inferred from a dramatic image or a category name. The same movement deficit can arise through different mechanisms, while one mechanism can produce different observed outcomes depending on body state and context. A useful model must retain enough of that context to distinguish the possibilities its data support.

We would begin with a declared prediction, a relevant comparator and a reserved test. Does adding a signal improve the estimate beyond uncertainty? Does the benefit persist under the population and conditions named in the claim? A model trained on one task or species does not inherit clinical validity in another. Differences which matter to the intended use belong in the evaluation rather than a footnote after promotion.

That is a narrower promise than an all-purpose AI doctor, and a more demanding one. The model has to make a useful difference which we can identify. Representation, context and grounding give us three connected ways to ask why it might succeed, and three places to look when it fails.

Reveal Cues

1. Show the title, image and subtitle together. Explain the parallel virtual humanoids, then distinguish data, a bounded prediction, deployment and evidence of clinical benefit.

References

Brain2Body programme ethos, current source

Dumb Medical Models

Slide 13 | Reveals 13-14

Dumb and smart are the rhetorical endpoints. The useful distinction is whether a medical model makes a specified prediction that can be checked, carries relevant context and remains reliable under the conditions where it is intended to be used. A large model is not automatically smarter in that practical sense.

A simple clinical rule can be useful and a complex model can fail. Compare the two on the same target, using appropriate held-out cases and reporting uncertainty, errors and population limits. Deployment is another test: a model that performs on a retrospective dataset has not thereby demonstrated clinical benefit.

Reveal Cues

1. Reveal Dumb Medical Models with its outgoing down arrow; frame the label as a rhetorical starting point.
2. Add Smart Medical Models; define useful performance through fair comparison, independent testing and limits.

References

Brain2Body programme ethos, current source

Deepblue To Deepmind

Slide 14 | Reveals 15

IBM’s Deep Blue defeated the reigning world chess champion in a match in 1997 using high-speed search and a chess evaluation system. DeepMind is an organisation; its AlphaGo system later combined deep neural networks with search for Go. These are different systems and tasks, not a controlled experiment proving that depth alone caused a general intelligence transition.

AlphaGo did not win by looking at a board photograph and declaring it plausible. A policy network estimated promising moves from a position. A value network estimated the likely outcome from that position. Search then examined possible continuations, using those estimates to concentrate effort where it could change the decision. Expert games and self-play supplied learning experience; the rules and the result supplied a task-specific test. Silver and colleagues' 2016 system made those components work together in Go.

Representation is the part of that story we need here. A raw observation is not yet a useful variable for every decision. A model can learn features which preserve the relationships needed for a prediction while discarding other detail. Depth gives the network a way to compose features; it does not guarantee that those features correspond to the biological mechanism we hoped to find.

In a neural recording, a useful feature might track a task event, a movement tendency or a recording artifact. They can all reduce training error. Inspecting a learned representation and testing it under appropriate changes helps distinguish those explanations. Predicting reserved data is necessary, and the unit held out must match the claim: a new trial is not automatically a new subject or a new task.

The analogy therefore has limits. Go provides explicit rules and a clearly scored outcome. A biological recording has measurement uncertainty, hidden conditions and competing explanations. We want AI's capacity to learn useful structure, while making the experimental task and independent reference explicit enough that a successful representation means something.

Reveal Cues

1. Show the title, image and subtitle together. Distinguish IBM’s Deep Blue from DeepMind’s AlphaGo and retain the task-specific comparison.

References

IBM, Deep Blue https://www.ibm.com/history/deep-blue

Silver et al. (2016), Mastering the game of Go with deep neural networks and tree search https://pubmed.ncbi.nlm.nih.gov/26819042/

Brain2Body programme ethos, current source

Representation

Slide 15 | Reveals 16

Deep learning builds a representation through successive learned transformations. Earlier features can be combined into relationships useful for the model's training objective. That makes it possible to work with complex inputs without specifying every feature by hand.

The learned representation is useful only for a stated test. In a neural recording it could preserve information about a task, but it could also capture movement contamination or a recording artefact. Hold out the relevant subjects, sessions or tasks, compare simpler baselines and state which observations were actually available.

Depth is a computational design choice, not proof that the model has discovered a biological mechanism. Structure, measured activity and the body's independent outcome still have different jobs in the explanation.

Reveal Cues

1. Show the title, image and subtitle together. Explain learned representation and the independent test needed to distinguish useful features from artefacts.

References

Google DeepMind, AlphaGo https://deepmind.google/research/alphago/

Brain2Body programme ethos, current source

IBM Watson To AlphaGo

Slide 16 | Reveals 17

IBM's Watson DeepQA answered questions in a televised Jeopardy! contest in 2011 by generating and evaluating candidate answers against evidence. AlphaGo addressed a different problem: selecting moves in Go using deep neural networks and search. Both had defined tasks and ways to score their outcomes.

The comparison is a way to discuss changing methods, not a head-to-head benchmark or a claim that one system directly succeeded the other. A question-answering contest and a board game have different rules, data and errors. For our work, the transferable lesson is to specify what the model receives, what it must predict and how the answer can be checked.

Neural and body measurements add uncertainty that a game board does not. We need independent movement or physiological outcomes, appropriate baselines and tests on genuinely reserved conditions. A memorable AI milestone does not establish biological or clinical performance.

Reveal Cues

1. Show IBM Watson To AlphaGo and its illustration. Identify the different tasks and connect them to a checkable prediction.

References

IBM Research, Building Watson: An Overview of the DeepQA Project https://research.ibm.com/publications/building-watson-an-overview-of-the-deepqa-project

Google DeepMind, AlphaGo https://deepmind.google/research/alphago/

Brain2Body programme ethos, current source

Autocomplete To Attention

Slide 17 | Reveals 18

Autocomplete predicts a continuation from available context. Attention is a mechanism for relating elements within a sequence; the two are not mutually exclusive, because a Transformer can itself be used to predict text. The title marks a change in tools and ambition, not a clean boundary between two unrelated capabilities.

A word changes meaning with its neighbors. A neural observation also arrives in a situation: a particular body, task, stimulus, time and recent history. Context is the information that makes a relationship interpretable, not merely a longer recording.

Transformer attention supplies a computational mechanism for weighting relationships among sequence elements. In simplified form, attention compares learned queries and keys, scales their scores and uses the resulting weights to combine values. The network learns what those transformations should preserve for its training objective. Vaswani and colleagues introduced that architecture for sequence tasks; attention weights are not a direct picture of a person's awareness or a proof of biological causality.

For Brain2Body, context begins before the model. Sensors must refer to the same event on a shared, checked clock. A sequence model cannot repair an unknown time offset simply by being large. It may instead learn a shortcut tied to that offset. The body pose, task conditions and provenance of each stream must travel with the observations.

Going from narrow to general is an ambition to test across specified changes. It has no automatic endpoint where context makes a model universally competent. We should ask which new conditions it handles, which observations it needed and where performance falls. Attention can model relationships the experiment makes available. It cannot supply missing measurements or make incompatible recordings comparable by declaration.

Reveal Cues

1. Show the title, image and subtitle together. Explain autocomplete and attention as compatible ideas, then qualify the ambition to generalise.

References

Vaswani et al. (2017), Attention Is All You Need https://arxiv.org/abs/1706.03762

Brain2Body programme ethos, current source

Context

Slide 18 | Reveals 19

A Transformer uses attention to combine information across elements of a sequence. In the original architecture, attention relationships are learned for the training objective. This gives a model a way to use context; it does not guarantee that the context is correct or sufficient.

For brain-body data, context has to be assembled before the model sees it: the relevant body, task, sensor identity and aligned time. A clock error or a measurement taken after the predicted event can become a shortcut. A model with a long context can therefore be confidently wrong about which events belong together.

Test generalisation on the changes the claim actually names. A result on another trial is not automatically a result on another person, task or species. Record the permitted inputs, the held-out unit and the errors.

Reveal Cues

1. Show the title, image and subtitle together. Explain attention as a way to use context, then identify the measured context and held-out test.

References

Vaswani et al. (2017), Attention Is All You Need https://arxiv.org/abs/1706.03762

Brain2Body programme ethos, current source

Google Translate To ChatGPT

Slide 19 | Reveals 20

Google Translate and ChatGPT make a change in language interfaces familiar. Google described its move to neural machine translation in 2016. OpenAI introduced ChatGPT as a conversational system in 2022. Translation and conversation overlap in possible uses, but they are not the same task or a controlled comparison of general intelligence.

The question for this programme is not whether a neural recording has a hidden sentence waiting to be translated. It is whether a declared relationship between observations and an independently measured outcome can be learned in a stated task. Different people, bodies and recording conditions can make the relationship ambiguous.

Test what information survives the mapping, report uncertainty and compare with simpler alternatives. A convincing sentence or animation is not evidence that subjective experience, intention or a retained skill has been copied.

Reveal Cues

1. Show Google Translate To ChatGPT and its illustration. Distinguish language products from a tested brain-body mapping.

References

Google Research, A Neural Network for Machine Translation, at Production Scale https://research.google/blog/a-neural-network-for-machine-translation-at-production-scale/

OpenAI, Introducing ChatGPT https://openai.com/index/chatgpt/

Brain2Body programme ethos, current source

GPT-3 To O3

Slide 20 | Reveals 21

GPT-3 is a language model trained autoregressively; o3 is a later reasoning model. An agent is a system arrangement that may use a model with tools and observations, not a guarantee supplied by a model name. Reasoning or tool access alone does not establish that a result has been independently verified.

A chatbot can say it marked the paper. An agent can open the paper, change a mark and inspect the result. The distinction becomes useful only when something outside the explanation can establish whether the intended work happened.

ReAct is one published example of combining language-model reasoning with actions and external observations. Its authors evaluated specified question-answering and interactive tasks. That work supports the value of acting and observing in those settings; it does not establish that every agent is grounded, reliable or generally intelligent.

Marking papers also exposes the weakness of a convenient reward. An answer can match a marking key while missing the question's intended meaning. A system can optimize the visible score without improving the thing the score was supposed to measure. The external check must therefore correspond to the actual target, and the model must not be allowed to choose its own final exam.

Our equivalent is independent movement measurement. A predicted trajectory is compared with a calibrated reference on reserved observations, in aligned time and physical units. A physics engine adds useful constraints, but its output is still a model's verdict. Matching a simulator is not the same evidence as matching biology. Grounding means making the claim answerable to the relevant outside outcome, with the measurement's uncertainty intact. It gives the model a disciplined chance to be wrong, which is exactly the chance we need.

Reveal Cues

1. Show the title, image and subtitle together. Distinguish model names from an agent workflow and identify an independent outcome.

References

OpenAI, Language Models Are Few-Shot Learners https://openai.com/index/language-models-are-few-shot-learners/

OpenAI, o3 model documentation https://developers.openai.com/api/docs/models/o3

Yao et al. (2023), ReAct: Synergizing Reasoning and Acting in Language Models https://arxiv.org/abs/2210.03629

Brain2Body programme ethos, current source

Grounding

Slide 21 | Reveals 22

Reinforcement learning can use feedback from a defined reward to improve a policy. Verification asks whether a result satisfies a specified check; some systems use such checks to supply rewards. A check can miss what the real task requires, so a training reward and an independent test of the intended outcome remain different.

Verification asks what observation could show the model is wrong. An agent's own answer, a training score and an independent outcome are different things. If a model predicts a reach, compare that prediction with a calibrated movement measurement on data reserved from selection.

The comparison must be fair. Keep the same target, units, prediction horizon and allowed information for the candidate and simpler baselines. Report timing errors, uncertainty and cases that fail, not just the favorable average. A physics engine can rule out some implausible movements but is itself a model, not an independent biological measurement.

Grounding connects the model's claim to a relevant outside consequence. It does not turn correlation into cause, make a private experience observable or establish a clinical treatment. Each stronger claim needs the evidence appropriate to it.

Reveal Cues

1. Show the title, image and subtitle together. Explain how a check can inform reinforcement while a calibrated independent outcome grounds the claim.

References

Brain2Body programme ethos, current source

Helff et al. (2026), LLMs Gaming Verifiers: RLVR can Lead to Reward Hacking https://arxiv.org/abs/2604.15149

Task Augmentation To Automation

Slide 22 | Reveals 23

Task augmentation gives a person information or assistance while the person remains involved in carrying out a task. Automation assigns a specified part of the work to a system. Either can be useful; the name alone does not tell us who made a decision, what was measured or whether the outcome improved.

For a proposed motor task, an agent might suggest a correction, guide a virtual action or perform part of the action itself. Those are different roles. Record what the person and the system actually control, compare against an appropriate baseline and score the same independently measured outcome.

An assisted result is not automatically a learned skill. If the question is retention, test performance after trainer-specific information stops, preserving any necessary compensatory bridge. A successful automated or simulated action does not by itself establish biological skill transfer.

The two photographs show a human–robot collaborative sawing experiment and an industrial robot production line. They illustrate different relationships between people and machines; neither establishes that this programme's proposed system works.

Reveal Cues

1. Show the title and both side-by-side photographs. Distinguish the collaborative sawing experiment from industrial automation and from the proposed system.

References

Brain2Body programme ethos, current source

Luka Peternel, human–robot collaborative sawing, CC BY-SA 4.0 https://commons.wikimedia.org/wiki/File:Human-Robot-Collaboration-Sawing-2016-Luka-Peternel.jpg

Clemenspool, industrial robot production line, CC0 https://commons.wikimedia.org/wiki/File:Staeubli_Robots_line.jpg

REPRESENTATION

Slide 23 | Reveals 24-26

Representation: what did we preserve? A recording is already a selection. Electrode position, sampling, filtering and feature extraction determine which changes enter the model. A learned representation then selects further relationships useful to its objective. We need to know whether the useful feature belongs to the intended task or to an accidental shortcut.

Context: which event does it belong to? The same measured feature can accompany different actions in different states. Keep task conditions, body pose, sensor identity and time alignment attached. If we confuse two trials, a model may become better at predicting the confusion. More computation cannot establish that observations came from the same event.

Grounding: what outside observation can reject the interpretation? Movement gives us position, timing, contact and task outcome that can be measured independently, within the limits of the instrument. Reserve the relevant observations before model selection, and test against simpler accounts. A held-out trajectory cannot tell us everything happening inside a brain. It can tell us whether this particular prediction of the trajectory was wrong.

The arrows connect dependencies rather than a guaranteed ladder of progress. A poor representation may discard the needed variable. A well-represented signal may lack its context. A richly contextual model may still answer only to its own score. The three words become useful when we can point to the observation, the condition and the external test behind each one.

Reveal Cues

1. Reveal REPRESENTATION with its following arrow; discuss the information retained.
2. Add CONTEXT with its following arrow; preserve the first item and connect task/time conditions.
3. Add GROUNDING; identify independently measured movement and its limits.

References

Brain2Body programme ethos, current source

For End-To-End Emulation

Slide 24 | Reveals 27

A knot is made by bringing the path back through itself. A sensorimotor loop closes when an action changes what is sensed and that new information can affect the next action. Joining software boxes with arrows is a proposal for that path; observing the return is how we establish it.

There are several loops we could mean. An offline prediction compares a recorded output with a recorded reference. A simulated controller acts in a model environment and receives simulated feedback. A biological loop involves an actual organism whose changing state is measured. The same mathematical notation may describe all three, but their evidence and sources of error differ.

For an end-to-end test we would specify the starting observation, transformation, action or prediction, independent outcome and return signal. Timing belongs in every link. Delays can change the control problem even if every individual component appears accurate when tested separately. Calibration and uncertainty also propagate: an error at the movement reference can be mistaken for an error in the neural mapping.

Closing the loop is therefore a testable property of a particular arrangement, not a synonym for complete emulation. We must show which links ran, what information each used and what failed under reserved conditions. If a component was prescribed, replayed or supplied by a hand-written controller, say so. The whole result belongs to that arrangement until independent biological evidence warrants a more specific claim.

Reveal Cues

1. Show the title, image and subtitle together. Explain the image and subtitle together; develop the mechanism and its limit.

References

Brain2Body programme ethos, current source

DNA

Slide 25 | Reveals 28-30

DNA supplies a template. During transcription, cellular machinery copies a selected stretch into RNA; the resulting transcript can be processed before it is used. During translation, a ribosome reads messenger-RNA codons and assembles an amino-acid chain. Folding, location and interaction then matter to what the protein does. This starting chain describes protein-coding expression, rather than a complete flowchart of a living cell.

An RNA count cannot, on its own, tell us how much active protein a cell contains. Transcripts differ in stability and translation; proteins differ in degradation, processing and activation. Some RNAs function without becoming proteins at all. Proteins also help regulate processes upstream. Feedback adds dependencies; it does not mean that arbitrary sequence information flows backward through every arrow. Each step therefore needs observations of its own, in the relevant cell and conditions.

Why spend time on this? Because the gap between a description and a consequence lives inside those relationships. A DNA change may have little effect in one context and matter in another. Measuring an RNA abundance does not directly measure every protein's abundance, activity or interaction. Predicting a shape is another valuable observation, with another target.

A connected model would have to state the quantities crossing its interfaces and the conditions under which each relationship was learned. Then it would need comparisons with independent measurements at the relevant stages. Displaying the chain does not establish those connections. It makes the missing tests visible, which is the point we carry from genomics into the nervous system.

Reveal Cues

1. Show DNA and its down arrow; distinguish sequence from activity.
2. Add RNA and its down arrow; explain expression, processing and noncoding scope.
3. Add PROTEIN; explain why structure and living consequence still require tests.

References

NHGRI, Talking Glossary of Genomic and Genetic Terms https://www.genome.gov/genetics-glossary

Brain2Body programme ethos, current source

AlphaGenome

Slide 26 | Reveals 31-33

AlphaGenome takes DNA sequence and predicts specified functional genomic measurements. Avsec and colleagues' 2026 paper includes expression, splicing and several regulatory readouts. It already crosses the tidy boundary between the DNA box and the RNA box. That is useful progress toward connecting sequence with measured consequences, not a complete model of a cell's life.

AlphaTranscriptome names a desired RNA and cell-state prediction capability. It remains a proposed programme component: a model relating specified RNA observations and context to declared cellular targets. It is not a released product being demonstrated here. Existing work such as scGPT has evaluated single-cell representations on defined tasks, including annotation and perturbation-related predictions. Those precedents motivate the question without establishing every cell process or a universal state-to-function model.

AlphaFold 2 supplies a different success: predicting protein structure, tested against experimentally determined structures in CASP14. Information in the amino-acid sequence and in related sequences helps the system infer geometric relationships, which are developed into a three-dimensional prediction. A predicted fold can guide a functional hypothesis. It does not directly measure all of the molecule's dynamics or its consequence inside a living system.

The important object is the relationship between the boxes. Their inputs and outputs have different units, resolutions, coverage and assumptions. Naming three models does not make their outputs compatible, connect their errors or demonstrate an end-to-end loop. We would have to define those interfaces, validate the linked predictions and expose what remains missing. The progress is real; the closed living-cell emulator is not a result this presentation can claim.

Reveal Cues

1. Reveal AlphaGenome with its arrow; explain its DNA input and genomic targets.

2. Add AlphaTranscriptome with its arrow; explicitly identify the intended AlphaTranscriptome as proposed.

3. Add AlphaFold; explain measured structure benchmarks and the unverified integration between components.

References

Avsec et al. (2026), Advancing regulatory variant effect prediction with AlphaGenome
https://pmc.ncbi.nlm.nih.gov/articles/PMC12851941/

Cui et al. (2024), scGPT: toward building a foundation model for single-cell multi-omics using generative AI
https://www.nature.com/articles/s41592-024-02201-0

Jumper et al. (2021), Highly accurate protein structure prediction with AlphaFold
https://pubmed.ncbi.nlm.nih.gov/34265844/

An End-To-End Pipeline

Slide 27 | Reveals 34

An end-to-end pipeline connects a declared starting input to a declared output through named transformations. For the molecular analogy, the stages may use different data, targets and assumptions. Their names placed in sequence do not establish that they are compatible, integrated or validated as a complete biological system.

The AlphaTranscriptome label in this presentation is a proposed component. AlphaGenome and AlphaFold address different tasks; an actual combined pipeline would need specified interfaces, compatible inputs and outputs, and tests against reserved measurements. A pipeline can run from one end to another without having a feedback loop.

For emulation, ask which outcome the full arrangement reproduces, in what units and under which conditions. Report the error and what was supplied or omitted. The implementation earns its scope through the end-to-end test, not through the word pipeline.

Reveal Cues

1. Show the title, image and subtitle together. Describe the proposed molecular pipeline and distinguish end-to-end processing from demonstrated closed-loop emulation.

References

Brain2Body programme ethos, current source

A Closed Loop

Slide 28 | Reveals 35

How do we ask the same connected question of a nervous system? Begin where the output can be observed independently: a body moves, contacts an object or fails to complete a task. Then ask which recorded observations predicted that outcome and which missing relationships account for the error.

Neural structure supplies constraints. Activity supplies time-dependent observations. Muscles and body mechanics turn some of those processes into forces and motion. Sensory consequences change what arrives next. The proposed loop connects these observations without declaring that any one of them reveals a unique private intention or a complete hidden state.

End-to-end must have a declared end at each side. For an initial prediction study, that might be specified brain and muscle observations as input and a measured trajectory as target. For a later closed-loop study, we would also need a qualified output interface, measured consequences and the return signal. A diagram including all of these is not evidence that they have all operated together.

There is no reason to demand a universal nervous-system model before asking a useful local question. Choose a bounded task, freeze the comparison and reserve relevant data. Does a richer model improve the named prediction beyond a simpler baseline and the measurement's uncertainty? If it fails, identify where and keep the failure. A concrete loop gives us a way to accumulate explanations, instead of treating every new map as the finished organism.

Reveal Cues

1. Show the title, image and subtitle together. Explain the image and subtitle together; develop the mechanism and its limit.

References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

Start With Movement

Slide 29 | Reveals 36

Start with movement because physical consequences can be checked. The swimming-pool meme makes the stakes stark: breathing and escaping danger involve bodily action. Yet the subtitle must not be read as a claim that the inability to move one's limbs means imminent death, or that disability diminishes a person's life.

Respiration, circulation, voluntary movement and assisted function involve different mechanisms and supports. A visible action is only one part of what keeps an organism alive. Our argument needs no universal no-movement rule. It needs a defined output whose success, failure and timing can be measured independently of the system predicting it.

A controlled task might ask whether a body reaches a target or navigates a declared path under stated conditions. Position alone may be insufficient. A path can be geometrically correct and late, or reach the endpoint with an unsupported contact. We would specify which dimensions matter before evaluating the candidate and report uncertainty in those measurements.

The survival imagery supplies the reason to care, not an experimental endpoint we are proposing to induce. The actual research question remains bounded: can synchronized observations support a useful prediction or, later, a qualified feedback relationship? Begin with a task we can measure well. Preserve its limits. A broad aspiration becomes scientifically tractable through a small outcome which can decisively show the proposed account was wrong.

Reveal Cues

1. Show the title, image and subtitle together. Explain the image and subtitle together; develop the mechanism and its limit.

References

Brain2Body programme ethos, current source

Movement Is The Ultimate Problem

Slide 30 | Reveals 37

Movement is where many biological demands become an observable action. An animal has to reach food, avoid danger, find a mate or adjust its position. A raccoon stuck in a drain grate makes a movement constraint immediately recognizable; it does not reduce every nervous process to visible locomotion.

The advantage for this programme is experimental. A trajectory has physical units and timing. Contact can occur or fail to occur. A task can be defined before the result is seen. These observations give a neural prediction an external target, rather than asking the model to agree only with an interpretation supplied by the same model.

The ultimate problem is therefore a choice of starting point, not a ranking that dismisses memory, emotion, internal sensing or people with restricted movement. The nervous system also regulates and represents processes which are not captured by a visible limb trajectory. Even in a movement task, knowing the hand's path does not reveal the person's unique intention. Several intentions and control strategies can produce a similar reach.

We can still use movement to constrain an explanation. Add muscle activity to specified brain observations; ask whether it improves a held-out prediction. Include body mechanics; ask whether it excludes an otherwise plausible but impossible trajectory. Retain the task and sensory context. Each measured relationship narrows the possibilities. That is enough reason to begin here, without claiming that the first target exhausts the biology we eventually want to understand.

Reveal Cues

1. Show the title, image and subtitle together. Explain the image and subtitle together; develop the mechanism and its limit.

References

Brain2Body programme ethos, current source

Whole-Brain Emulation

Slide 31 | Reveals 38

A brain belongs to a body. Its outgoing activity meets muscles, joints and loads; its incoming observations depend partly on what that body did. A whole-brain ambition which omits those relationships leaves part of the functional question outside the model.

Whole-body emulation is the corresponding ambition, not a claim that every molecule must be explicitly simulated. We would need the parts relevant to the declared behavior, their coupling and the boundaries where unmodelled processes enter. A simplified body can be useful if it preserves the movement and feedback relationships being tested. A detailed body can still be wrong about forces, contact or sensory access.

The nervous-system view must distinguish anatomy from function. Afferent pathways carry information toward the central nervous system, and efferent pathways away from it; those physiological directions are different from input and output relative to a computer. The spinal cord also contains local processing, so the body is not a passive cable waiting for a complete command from the head.

A credible first step would test whether adding body observations improves a named prediction beyond a brain-only baseline. We should show where the added information helps, where it does not and what remains unobserved. That result would earn its bounded claim. A lifelike nervous-system model is a valuable teaching object; it is not, by its appearance, a validated whole-body emulator.

Reveal Cues

1. Show the title, image and subtitle together. Explain the image and subtitle together; develop the mechanism and its limit.

References

Brain2Body programme ethos, current source

Virtual Molecules

Slide 32 | Reveals 39-40

Virtual molecules and virtual cells are descriptions at different physical scales. They may also use different computational levels of detail. Those two choices should not be confused: a model can describe a large physical object with few variables or a tiny component with many.

A molecular model might predict a structure or a specified interaction. A cell model might predict a measured state, response or change over time. Passing information from one to the other requires a defined interface: what quantity, in what units, for which cell type and conditions, with what uncertainty? A colorful animation cannot supply those missing definitions.

Nor do we gain a complete cell by collecting every molecular prediction. The arrangement, concentrations, localization and interactions matter; observations constrain only some of them. A useful reduced model might describe the outcome without computing every molecule individually. We should choose the level of detail from the question and the available evidence, then compare the resulting prediction with an independent reference.

This protects us against two shortcuts: treating microscopic detail as automatic truth, and treating a good aggregate fit as proof of every local mechanism. Statistical averaging is another axis again. An average across cells may conceal different responses in individuals without saying anything about the physical resolution of the model. The down arrow therefore marks a proposed relationship to characterize, rather than a declaration that the scales have already been unified.

Reveal Cues

1. Reveal Virtual Molecules with its following down arrow; define a physical scale and a target.
2. Add Virtual Cells; explain the proposed interface and independent tests in each direction.

References

Brain2Body programme ethos, current source

Virtual Bodies

Slide 33 | Reveals 41-42

Virtual bodies and virtual brains also need an interface we can inspect. A rendered brain beside a rendered animal is a composition. A functional connection requires declared observations, control variables, dynamics, sensory consequences and time alignment.

For the body, choose the mechanics the task needs. Joint geometry, mass distribution, muscle action and contact can constrain movement; each remains an assumption or an estimate until supported by relevant measurements. Established musculoskeletal tools such as OpenSim provide ways to express and analyze such models. Access to a simulator is not evidence that our chosen animal, parameters or controller have been validated.

For the neural side, distinguish a connectivity constraint, an activity observation and a fitted dynamical rule. They can enter one model without becoming the same kind of evidence. A predicted control variable must have a defined route into the body model, and the returning observation must correspond to what the organism or recording could actually access.

Then test each direction. Can specified observations predict the movement? Can the movement and context constrain a named recorded feature? The second question does not inherit a unique answer from the first. Multiple hidden states can generate similar motions. Keep uncertainty where that ambiguity remains. The aim is a coupled account whose predictions survive outside the fitting data, not two convincing models which silently certify one another.

Reveal Cues

1. Show Virtual Bodies with its following down arrow; identify mechanical and measurement assumptions.

2. Add Virtual Brains; explain control/feedback interfaces and separately tested directions.

References

Seth et al. (2018), OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1006223

Brain2Body programme ethos, current source

Brains REPRESENT Environments.

Slide 34 | Reveals 43-45

Brains represent environments in the practical sense that neural activity can carry information useful for acting in a situation. The statement does not mean the head contains a literal miniature of the room, or that every recorded feature has one fixed meaning.

Think of estimating where your hand is when you can no longer see it. Sensory information and the consequences expected from an action can contribute to an estimate. Wolpert, Ghahramani and Jordan tested a constrained version of that problem in 1995, using hand-location judgments under specified movement and force conditions. Their result supports an internal-model account in that task, rather than a complete theory of all representation.

For our study, a representation needs a stated target. If a feature predicts the next turn, what observation would distinguish a spatial relationship from a familiar trial sequence? If it predicts position, does it preserve timing and uncertainty? A decoder's success shows that information useful to its test was available; it does not uniquely reveal the code used by the organism.

This is why the environment must be described alongside the recording. The available cues, task history and sensory limits affect what the observation can mean. A model that answers those conditions accurately has earned a bounded prediction. Calling its internal variable a world representation cannot expand that result to every unobserved situation.

Bodies give neural activity its immediate context. The same intended reach meets a different problem when the elbow is already bent, the object is heavier or the body is fatigued. A recording detached from those conditions loses information about the action it accompanies.

The mechanical side is easy to grasp. A joint's position changes leverage and available movement. The physiological side adds further unobserved state. We should measure the variables relevant to the declared question where possible and distinguish those observations from fitted estimates. Surface muscle activity, for example, is an electrical observation; it is not a direct readout of every force or sensory signal.

Context does not give us a unique inverse. A particular trajectory may be produced by different muscle combinations and control strategies. Adding body observations can narrow the explanations while leaving several possible accounts. A model should express that uncertainty rather than returning a falsely exact hidden state because its interface prefers one number.

Our test is whether the extra context improves a prespecified result on reserved observations. Compare with a simpler model supplied the same prediction horizon and permissible information. Watch for leakage from measurements taken after the event being predicted. If the body measurements help only because they already contain the answer, that is a different claim from forecasting. The body makes the question better defined; the experiment still has to establish the benefit.

An environment grounds a prediction when something there can contradict it. The hand either reaches the specified target within the allowed time or it does not, subject to the uncertainty of the measuring instrument. That outside comparison makes the claim answerable to more than its own internal story.

The relevant world is more than a background image. It supplies geometry, loads, contacts, available cues and consequences. A simulated agent given an exit's hidden coordinates faces a different task

from an animal which must discover the route through its permitted senses. A fair account records the information available to each and the controller that transforms it.

We should also distinguish a simulator's constraint from an independent biological reference. Physics can rule out some impossible motions, while an inaccurate contact model can create new shortcuts. Reward can favor the wrong outcome even in an otherwise adequate environment. Checking task completion, physical plausibility and biological agreement therefore requires separate evidence.

The environment does not solve private experience. Two people can make the same successful reach for different reasons. What it provides is a common observable consequence which can constrain our explanations. Keep the measurement calibrated, the final test reserved and the failure cases visible. Then a grounded prediction can make a precise contribution without claiming to have captured everything the brain experienced.

Reveal Cues

1. Reveal Brains REPRESENT Environments with its outgoing down arrow; define a testable prediction.

2. Add Bodies CONTEXTualise Brains with its outgoing down arrow; identify the relevant body state.

3. Add Environments GROUND Brains; identify an independent measured outcome.

References

Wolpert, Ghahramani and Jordan (1995), An internal model for sensorimotor integration
https://pubmed.ncbi.nlm.nih.gov/7569931/

Brain2Body programme ethos, current source

REPRESENTATION

Slide 35 | Reveals 46

The three terms now appear together. Representation asks what information the model retains; context identifies the body, task and timing to which that information belongs; grounding makes the prediction answerable to an independent outcome.

For a proposed movement test, those questions stay connected. A useful feature can still describe the wrong trial, and a well-aligned prediction can still fail against measured movement. Define the observations, preserve the relevant conditions and compare the result with an independent reference. None of the three labels alone establishes biological function or a clinical result.

Reveal Cues

1. Show REPRESENTATION, CONTEXT, GROUNDING and both arrows together immediately; connect them to the three preceding statements.

References

Brain2Body programme ethos, current source

Kung Fu THE HARD WAY

Slide 36 | Reveals 47

Kung fu the hard way means keeping the part the movie cut out. A nervous system has to receive information, act, encounter consequences and change in a way that can be tested. Our ambition is to understand a bounded version of that process, beginning with measurements that can challenge the model.

The big vision joins a proposed organism platform, specified human-related features, an interpretable measuring instrument, a separately qualified way of delivering information and, eventually, a test of learning. These are dependencies with different risks and different evidence requirements. No single demonstration can stand in for all of them.

We can obtain useful results along the way. A calibrated measurement that exposes a previously hidden relationship would matter. A model that improves a held-out prediction would matter. Assistance which enables an action would matter too. For a learning claim, however, the recipient must show an improvement beyond trainer-specific commands, with practice and necessary support handled fairly, and with retention or transfer tested as appropriate.

The dramatic endpoint should make those earlier questions easier to see, rather than make every partial result look like failure. We do not have to claim a universal neural language to ask whether a particular mapping works. We do have to identify the task, source, interface and outcome. Here is the roadmap, with every bold line tied to a question that can remain unanswered.

Reveal Cues

1. Show the title, image and subtitle together. Explain the image and subtitle together; develop the mechanism and its limit.

References

Brain2Body programme ethos, current source

1. Engineer The World's Greatest Model Organisms Platform

Slide 37 | Reveals 48

First: engineer the world's greatest model organisms platform. Greatest is the ambition, not a comparative result. Which organism, developmental stage and apparatus would expose the neural-body relationship we want to study without removing the biology that makes it matter? The platform would need repeatable observations, species-appropriate anatomy, known apparatus effects and independent outcomes. An accessible nervous system is not automatically a faithful model of every human condition.

Reveal Cues

1. Show numbered item 1; explain the platform and unmeasured superlative.

References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

2. Turn That Organism Human

Slide 38 | Reveals 49

Second: turn that organism human. Literally, we cannot. The scientific version is to study a specified human-related genetic feature or tissue contribution in a model organism and test what it changes. A selected feature, a viable graft and human equivalence are very different claims. The following stages keep that scope explicit rather than allowing the headline to promise a human mind in another animal.

Reveal Cues

1. Show numbered item 2; narrow humanisation to specified features.

References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

3. Build The Blood Pressure Monitor For The Nervous System

Slide 39 | Reveals 50

Third: build the blood pressure monitor for the nervous system. The analogy asks for interpretable measurement: a named quantity, calibrated observations, a task and an account of error. There is no single established nervous-system number equivalent to blood pressure supplied here. The proposed instrument would need its own validated targets and reference comparisons.

Reveal Cues

1. Show numbered item 3; explain calibrated measurement and the absent universal nervous-system number.

References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

4. Build The Pacemaker For The Nervous System

Slide 40 | Reveals 51

Fourth: build the pacemaker for the nervous system. Now the question changes from reading to writing. Information would have to be delivered through a separately qualified interface and its consequence measured. A cardiac pacemaker supplies an analogy of purposeful intervention; it does not prove the feasibility or safety of arbitrary neural instructions.

Reveal Cues

1. Show numbered item 4; distinguish delivery from recording and retain separate interface qualification.

References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

5. Use It To Make You Dance

Slide 41 | Reveals 52

Fifth: use it to make you dance. Movement while a controller supplies commands can establish assistance. A skill retained when trainer-specific guidance ends needs a different experiment, with appropriate comparisons, later testing and reserved changes of conditions. We would keep necessary compensatory support unchanged so the test measures learning rather than removal of basic function. A working measurement would be useful before the later ambitions were achieved. So would assistance that lets someone act. We can value each result without borrowing an untested claim from the next.

Reveal Cues

1. Show numbered item 5; distinguish assistance, retention and transfer.

References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

STAGE 0: Brain2Body

Slide 42 | Reveals 53

Stage zero would test the relationship between brain and body before adding the later humanisation ambitions. Brain2Body names that programme. It does not mean we already possess a complete translation of neural activity or a universally valid animal model.

Reveal Cues

1. Show the stage title. Introduce its proposed research question; the named steps follow individually.

References

Brain2Body canonical glossary, revision 33

Cook et al. (2021), Postnatal development in a marsupial model, the fat-tailed dunnart https://www.nature.com/articles/s42003-021-02506-2

Brain2Body programme ethos, current source

0.1: NeuroTranslate

Slide 43 | Reveals 54

NeuroTranslate begins with a relationship we can state: which recorded neural features predict which measured body variables, in which animal and task? The reverse direction asks a different question and needs its own test. A fitted animation alone cannot tell us whether either mapping works in a living animal.

The artificial-womb concept is proposed developmental access, not a qualified Antechinus life-support system. The newborn rendering uses an unregistered marsupial proxy; the species and apparatus cannot be assumed equivalent. Species-specific staging, care, reference anatomy and independent physiological evidence would be required before making a biological claim.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

References

Brain2Body canonical glossary, revision 33

Cook et al. (2021), Postnatal development in a marsupial model, the fat-tailed dunnart https://www.nature.com/articles/s42003-021-02506-2

Brain2Body programme ethos, current source

0.2: Navigation

Slide 44 | Reveals 55

A straight-walled maze gives navigation a declared beginning, end and set of available cues. Position, time and progress are separate observations. Record the actual outcome and the conditions under which it occurred before asking whether neural or body measurements can predict it.

The maze view is a proposed apparatus. A simulated animal following a path demonstrates the simulation, not successful navigation by a living animal. Compare against simple route and cue-based baselines and reserve conditions that were not used to fit the mapping.

The proposed bidirectional LiDAR illustration uses the maze floor as the boundary: translucent red above the floor and translucent blue below it. A colour in a schematic is not a measurement. Sensor calibration, coverage and independently checked movement are separate questions.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

References

Brain2Body canonical glossary, revision 33

Cook et al. (2021), Postnatal development in a marsupial model, the fat-tailed dunnart https://www.nature.com/articles/s42003-021-02506-2

Brain2Body programme ethos, current source

0.3: NeuroRelay

Slide 45 | Reveals 56

NeuroRelay names the proposed information link. State what is sent, where it came from, what receives it, and the time delay. A transmitted signal can be recorded perfectly while contributing nothing useful to the recipient's task.

The test is the difference the information makes. Compare the declared link with a matched control, including shuffled or absent information where appropriate, while measuring the same independent outcome. The rendering does not show an established biological relay or learned skill.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

References

Brain2Body canonical glossary, revision 33

Cook et al. (2021), Postnatal development in a marsupial model, the fat-tailed dunnart https://www.nature.com/articles/s42003-021-02506-2

Brain2Body programme ethos, current source

0.4: Orientation

Slide 46 | Reveals 57

Turns make orientation part of the question. The animal's position, heading, movement history and access to cues can all affect the route it takes. A known outcome needs a known environment: changes to the maze or its cues must be recorded with the observations.

A model that succeeds on a familiar sequence may simply have learned that sequence. Reserve route conditions for a later comparison, measure the movement independently, and keep the illustrated winding maze distinct from a completed biological experiment.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

References

Brain2Body canonical glossary, revision 33

Cook et al. (2021), Postnatal development in a marsupial model, the fat-tailed dunnart https://www.nature.com/articles/s42003-021-02506-2

Brain2Body programme ethos, current source

0.5: NeuroRelay

Slide 47 | Reveals 58

The second NeuroRelay beat returns to the link in the richer orientation task. A result from the straight-walled maze would not establish that the same signal carries useful information when the task and available cues change.

Keep source, recipient, timing and comparison explicit. Measure performance with and without the relevant information under matched conditions, then report which conditions were actually tested. The proposed connection is not evidence of a working interface.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

References

Brain2Body canonical glossary, revision 33

Cook et al. (2021), Postnatal development in a marsupial model, the fat-tailed dunnart https://www.nature.com/articles/s42003-021-02506-2

Brain2Body programme ethos, current source

STAGE 1: Geno-Humanisation

Slide 48 | Reveals 59

Geno-humanisation asks a feature-specific question: could a defined genetic change make a model more informative about a particular human-related mechanism? The answer cannot be read from the word humanisation. It would have to be established for the chosen feature, organism and outcome.

The value of a human-related feature is a better test of a declared question. It does not make the whole animal human, confer human cognition or establish that a treatment would translate to people. We would need an appropriate comparator and independent molecular and functional evidence before making even the narrower claim. This stage stays a proposal. The bold name states where the question points; it cannot stand in for the answer.

Reveal Cues

1. Show the stage title. Introduce its proposed research question; the named steps follow individually.

References

Brain2Body canonical glossary, revision 33

NHGRI, Talking Glossary of Genomic and Genetic Terms https://www.genome.gov/genetics-glossary

Brain2Body programme ethos, current source

Jinek et al. (2012), A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity https://pubmed.ncbi.nlm.nih.gov/22745249/

1.1: Geno-Transplantation

Slide 49 | Reveals 60

Geno-Transplantation is the roadmap's label for a proposed human-related genetic change. The illustrated needle entering an antechinus is a conceptual depiction of a possible approach; it does not define a target, establish an edit or show that a model organism has been humanised.

A sequence change, its molecular consequence and the function it is intended to clarify are different observations. Each would need a suitable comparison and independent evidence. The scope is a specified feature in a specified organism, not a claim of human cognition or clinical translation.

Begin with a declared hypothesis about a particular variant or its expression and a functional consequence. Compare it with a matched model without that change, and measure the molecular effect independently from the functional outcome. A difference may depend on developmental stage and species context; neither a sequence edit nor expression alone establishes the proposed functional effect.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

References

Brain2Body canonical glossary, revision 33

NHGRI, Talking Glossary of Genomic and Genetic Terms https://www.genome.gov/genetics-glossary

Brain2Body programme ethos, current source

Jinek et al. (2012), A programmable dual-RNA-guided DNA endonuclease in adaptive bacterial immunity https://pubmed.ncbi.nlm.nih.gov/22745249/

STAGE 2: Xeno-Humanisation

Slide 50 | Reveals 61

Xeno-humanisation asks a different question: what would specified human-related donor cells or tissue do in a host organism? Xeno-transplantation refers to transplantation across species. It is distinct from changing a sequence already present in a host.

The proposed experiment would need a defined cell source, host context, comparison and independently observed outcome. Species and developmental differences stay with the claim. No grafting study is performed by this model; any biological study would need its own institutional review and species-appropriate welfare and safety evidence. The aim is a more informative model of a selected relationship, with honest gaps between tissue presence, useful integration and broader human relevance.

Reveal Cues

1. Show the stage title. Introduce its proposed research question; the named steps follow individually.

References

Revah et al. (2022; updated article with 2026 Author Correction), Maturation and circuit integration of transplanted human cortical organoids https://www.nature.com/articles/s41586-022-05277-w

Brain2Body canonical glossary, revision 33

Brain2Body programme ethos, current source

2.1: Xeno-Transplantation

Slide 51 | Reveals 62

Xeno-Transplantation asks what a specified donor tissue contributes in a host. An organoid is a laboratory-grown tissue model; an assembloid combines selected components to study their interaction. Neither term means a complete human brain.

A graft might survive without creating the intended connection or changing a defined function. Tissue survival, anatomy, activity and an independent outcome therefore need distinct evidence. Published rat work is a limited precedent, not an Antechinus result. The image depicts a blade carrying tissue into an opening in an illustrated antechinus brain. The depicted fit and function are schematic and unvalidated.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

References

Revah et al. (2022; updated article with 2026 Author Correction), Maturation and circuit integration of transplanted human cortical organoids https://www.nature.com/articles/s41586-022-05277-w

Brain2Body canonical glossary, revision 33

Brain2Body programme ethos, current source

Revah et al. (2026), Author Correction: Maturation and circuit integration of transplanted human cortical organoids https://www.nature.com/articles/s41586-026-10943-4

Neural Composite

Slide 52 | Reveals 63

Neural composite: the proposed account joining declared neural observations, structural constraints and mapping components. Composite means that sources are combined with their identities and limitations intact. It does not turn recordings from different animals, tasks or instruments into one observed nervous system.

The artificial nervous system represents activity passing between sensing, central processing and movement. The composite would be built and tested against measured signals and actions: proposed connections are compared with independent observations and refined as needed. Different structures may produce similar behaviour, so function can constrain candidate structure without uniquely determining it. The illuminated pathways are schematic, not measurements or a validated simulation.

Reveal Cues

1. Show the title and conceptual artificial nervous system with activity across sensory, central and motor pathways.

References

Brain2Body programme ethos, current source

Seth et al. (2018), OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1006223

AI-generated conceptual illustration of an artificial nervous system

Physics Engine

Slide 53 | Reveals 64

Physics engine: the part which calculates mechanical consequences under stated assumptions. It needs variables with defined units, coordinate frames and time steps. A controller's output must have a meaningful route into that calculation. Joint limits, contacts and energy constraints can reject some impossible behavior, while bad assumptions or numerical artifacts can create convincing shortcuts. The rendered motion and the mechanical calculation also need to agree.

This AI-generated image illustrates exposed muscle. It is neither a simulation result nor a validated anatomical model. Physical consequences must be checked against relevant measurements.

Reveal Cues

1. Show the title and AI-generated illustration of an ape’s musculature.

References

Brain2Body programme ethos, current source

Seth et al. (2018), OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1006223

AI-generated conceptual illustration of an ape with exposed musculature

World Model

Slide 54 | Reveals 65

World model: the environment in which those consequences acquire a task meaning. The available cues, objects, obstacles and feedback determine the problem the model actually faces. A hidden goal coordinate or a prewritten route changes that problem even if the final animation looks like navigation.

The proposed world model aims to simulate movement over time across many parallel in-silico possibilities; the ambition is hundreds. The AI-generated illustration uses motion ghosts to suggest parallel movement over time; it is not measured simulation data, an exact count, biological accuracy or a clinical outcome.

Reveal Cues

1. Show the title and AI-generated motion illustration. Distinguish the schematic ghosts from measured simulation or biological results.

References

Brain2Body programme ethos, current source

Seth et al. (2018), OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1006223

AI-generated illustration created for this presentation; not measured simulation data.

Neural Composite

Slide 55 | Reveals 66-68

Neural composite: the proposed account joining declared neural observations, structural constraints and mapping components. Composite means that sources are combined with their identities and limitations intact. It does not turn recordings from different animals, tasks or instruments into one observed nervous system.

Physics engine: the part which calculates mechanical consequences under stated assumptions. It needs variables with defined units, coordinate frames and time steps. A controller's output must have a meaningful route into that calculation. Joint limits, contacts and energy constraints can reject some impossible behavior, while bad assumptions or numerical artifacts can create convincing shortcuts. The rendered motion and the mechanical calculation also need to agree.

World model: the environment in which those consequences acquire a task meaning. The available cues, objects, obstacles and feedback determine the problem the model actually faces. A hidden goal coordinate or a prewritten route changes that problem even if the final animation looks like navigation.

The arrows show an intended dependency, not a completed stack. We would test component behavior and then the assembled route, carrying timing and uncertainty through the interfaces. Neural prediction, simulated mechanics and independently measured movement remain separate references. A good fit at the endpoint cannot uniquely validate every hidden component. To make the composite explanatory, we need comparisons which can reveal which added relationship improved the result and which one introduced the error. That is the difference between a persuasive world on screen and a model answerable to the world outside it.

Reveal Cues

1. Reveal Neural Composite with its down arrow; identify provenance and declared neural inputs.

2. Add Physics Engine with its down arrow; explain mechanical units, timing and constraints.

3. Add World Model; explain permitted information, task outcomes and end-to-end independent tests.

References

Brain2Body programme ethos, current source

Seth et al. (2018), OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1006223

Large Movement Models

Slide 56 | Reveals 69-70

A movement can look right for the wrong reason. An animated arm follows a path because somebody supplied that path. A predictive model has the harder job: given only the inputs it is allowed to receive, estimate what happens next, and face a measurement it did not choose. That is the useful ambition behind Large Movement Models.

Start with one declared action. Neural observations describe part of the nervous system's activity; muscle observations describe another part of the action; independently measured motion gives us an external result to explain. The virtual body's geometry, joint constraints and mechanics restrict which explanations are physically plausible. A learning model would connect those observations across time. Each contribution needs its own job, because feeding the answer into a model and celebrating the answer coming out would be a very expensive way to draw a circle.

Large Synaptic Models pushes the ambition further, toward models of interactions within nervous systems. The name does not tell us that individual synapses have been measured, reconstructed or simulated. Model size, physical resolution and biological coverage are separate questions. State which variables exist in the implementation, their units or computational meaning, the training tasks and the reserved tests on which a proposed model earns its accuracy.

Reveal Cues

1. Reveal Large Movement Models with its outgoing down arrow; define a declared movement prediction.

2. Add Large Synaptic Models; distinguish a proposed model name from evidence of synaptic coverage.

References

OpenSim documentation, How Inverse Kinematics Works
https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090047/How+Inverse+Kinematics+Works

STAGE 3: Brain2Brain

Slide 57 | Reveals 71

Brain2Brain asks whether a defined information link from one nervous system can help another perform a stated task. The participants, information, delivery route and independent outcome must be declared; a connection alone establishes neither shared experience nor retained learning.

Rao and colleagues reported a task-specific human link using decoded motor imagery and transcranial magnetic stimulation to elicit a touchpad response. This is a narrow precedent; it did not demonstrate retained skill transfer and does not establish the proposed animal system.

Reveal Cues

1. Show the stage title. Introduce its proposed research question; the named steps follow individually.

References

Rao et al. (2014), A Direct Brain-to-Brain Interface in Humans
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

3.1: Mind-Meld

Slide 58 | Reveals 72

Mind-Meld is the first proposed Brain2Brain encounter. What could one system tell another that helps with a specified task? Name the sender, the observed signal, the recipient, the delivery route and the outcome before assigning meaning to the connection.

A connection in a rendering is not a working interface. Compare task performance with the intended information, an appropriate control and no information. Improvement while the link is active would still need a separate test before it could be called retained learning.

The rendering depicts a proposed direct brain-to-brain wire between the animals; it does not establish that an interface works or that experience has been transmitted.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

References

Rao et al. (2014), A Direct Brain-to-Brain Interface in Humans
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

3.2: NeuroTwin

Slide 59 | Reveals 73

Here NeuroTwin denotes the proposed biological trainer. It does not imply genetic twins or a validated digital copy. As the trainer learns the maze, we would record a declared part of its neural activity alongside the cue, its choices and its independently measured movement, all on a shared clock. The purpose is to ask which recorded features carry useful information about this task—not to assume we have captured the animal's experience.

Those features would need calibration against known task events. A second animal has its own body, history and neural responses, so the same recorded pattern cannot simply be assumed to mean the same thing in both brains. The single animal in the maze illustrates the proposed trainer; it does not establish a functioning brain-to-brain interface.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

References

Rao et al. (2014), A Direct Brain-to-Brain Interface in Humans
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

3.3: NeuroTransfer

Slide 60 | Reveals 74

Now connect the trainer to the trainee. A proposed interface would record the trainer's activity, transform a specified feature into a time-locked stimulation pattern, and deliver that pattern through a separately validated interface to the trainee's brain. The trainee's movement and choice would be measured independently. The route, timing and information delivered all matter: recording a signal is not the same as writing it into another brain.

There is a narrower precedent. Pais-Vieira and colleagues recorded cortical activity from one rat and used intracortical microstimulation to convey task-relevant information to another rat trained to use it. Their two-choice experiments did not establish that a whole maze skill or a private experience had been transferred. For our proposed maze test, compare the intended signal against matched practice and an appropriate control signal; success during the link would still be assistance, not proof of learning that persists after it.

A direct brain-to-brain link from the trainer on the left to the trainee on the right is proposed. The enlarged cutaways show contacts on simplified brain models; their anatomical fit is unvalidated, and no transfer is demonstrated.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Identify the proposed left-hand trainer as sender and right-hand trainee as recipient; no transfer is demonstrated.

References

Rao et al. (2014), A Direct Brain-to-Brain Interface in Humans
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

Pais-Vieira et al. (2013), A Brain-to-Brain Interface for Real-Time Sharing of Sensorimotor Information
https://doi.org/10.1038/srep01319

3.4: NeuroTwin

Slide 61 | Reveals 75

The trainer recording could be repeated with the task, available cues and measurement methods held fixed. If the trainer, trainee or conditions change, those differences need to be declared before comparing the results. Compare the signal with independently measured choices and timing; a relationship found for one animal may not carry over to another.

The single animal in the maze illustrates the proposed trainer; it does not establish a functioning brain-to-brain interface.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

References

Rao et al. (2014), A Direct Brain-to-Brain Interface in Humans
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

3.5: NeuroTransfer

Slide 62 | Reveals 76

A trainee may perform better while the trainer’s signal is present because it has received useful guidance. To ask whether anything has been learned, remove the trainer-specific information and measure performance again after a delay, keeping any separately necessary compensatory support constant. Compare that result with a suitable matched control, including equivalent practice and available cues.

Then test a reserved task variation that was not used to tune the mapping. Predeclare what counts as success and keep the independent movement and choice measures separate from the training signal. A retained advantage would need to be demonstrated under those conditions; it would not by itself establish that the trainer’s full experience was copied.

A direct brain-to-brain link from the trainer on the left to the trainee on the right is proposed. The enlarged cutaways show contacts on simplified brain models; their anatomical fit is unvalidated, and no transfer is demonstrated.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Identify the proposed left-hand trainer as sender and right-hand trainee as recipient; no transfer is demonstrated.

References

Rao et al. (2014), A Direct Brain-to-Brain Interface in Humans
https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0111332

STAGE 4: Agent2Brain

Slide 63 | Reveals 77

Agent2Brain asks whether a computational agent can help a model animal perform a declared task through a specified feedback route. An independent outcome is needed; assistance during a task and retained learning are separate questions. Human work comes later and needs its own evidence.

The input could be a time-aligned observation of the task and the animal; the agent converts that information into a prediction or feedback through a declared channel. Compare it with matched practice and a simpler feedback rule, then measure the action independently. A useful assisted response does not establish what the animal retains when the agent is removed.

Reveal Cues

1. Show the stage title. Introduce its proposed research question; the named steps follow individually.

4.1: Mind-Merge

Slide 64 | Reveals 78

Mind-Merge opens the proposed agent-to-animal experiment. The agent observes a declared task and may produce a prediction or correction; any feedback needs an identified route the animal can receive. A diagram does not establish that route.

Measure the animal’s action, the information available to the agent, and what changes with feedback. A drawn connection does not prove that an agent can read or write thoughts.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

4.2: NeuroTrainer

Slide 65 | Reveals 79

NeuroTrainer is the proposed software-agent role. It might predict a missed target or a late movement, but the prediction must be tied to a declared observation and task. A persuasive explanation on a screen is not a measured correction.

Compare feedback against a simpler or matched alternative, on the same task and clock. Record the agent's available information so that a result cannot be credited to a neural model when it came from an unreported cue. The trainer shown here is illustrative.

For example, a predicted timing error could become an ordinary visual cue. Compare the same cue timing and practice exposure with a simple rule; reserve task variations to test whether the agent contributes more than access to information.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

4.3: NeuroTrainee

Slide 66 | Reveals 80

NeuroTrainee names the model animal receiving feedback and performing the task. The outcome is the animal’s independently measured action, not the agent’s prediction. The receiving interface and observations must be identified; a diagram does not establish a working neural writer.

Declare the feedback channel and compare performance under matched conditions. Improvement during guidance may be useful assistance, but does not show that learning remains without the guide.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

4.4: NeuroTrainer

Slide 67 | Reveals 81

NeuroTrainer appears a second time in the roadmap. The label by itself does not establish a second agent or a new architecture. It does give us a reason to revisit the trainer's contribution: which information did it supply, and what did the comparison show?

Keep the agent's information and the task exposure documented. The outcome can then be compared with a version that does not supply the trainer-specific help. The visual is a proposed setup, not a completed study.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

4.5: NeuroTrainee

Slide 68 | Reveals 82

The second NeuroTrainee entry takes us back to the learner. After practice, remove the trainer-specific information and measure whether the improvement remains, with an appropriate comparison and any necessary compensatory bridge kept constant.

A delayed test and an unused task variation ask more than an immediate assisted performance score. The result would still belong to the animals and conditions tested; the illustration does not establish retained skill or general transfer.

Track performance during guidance, immediately after guidance is removed, and later on a reserved variation. Those measurements distinguish a temporary aid from retention and generalisation; they are proposed outcomes, not results already obtained.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

STAGE 5: Agent Vs Neurology

Slide 69 | Reveals 83

Pick a disease, and the temptation is to pick a cure. The experiment has to begin further down to earth: which condition, which functional problem, which measurable change? Agent Vs Neurology proposes a way to compare candidate explanations and actions. It supplies no evidence that an agent can treat the chosen disease.

The proposed comparison begins with measured function, not a disease name alone. Feed declared observations into a model, specify the agent's accessible information and candidate action, and compare its predictions with a simpler baseline on reserved conditions. Simulation errors, missing measurements and reward shortcuts can all produce plausible but misleading results; biological benefit requires a separate independent test.

Reveal Cues

1. Show the stage title. Introduce its proposed research question; the named steps follow individually.

5.1: Pick Your Disease

Slide 70 | Reveals 84

Pick Your Disease begins with a specified condition and a functional problem, not with an assumed cure. Which observable outcome matters, for whom, and under what conditions? The answer sets the comparison the model must eventually face.

A simulated disease label cannot establish a biological mechanism or a clinical benefit. The proposed work needs appropriate measurements and independent tests before a model result can support either claim. The image is illustrative.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

5.2: Pick Your Agent

Slide 71 | Reveals 85

Pick Your Agent makes the computational role explicit. A predictor estimates an outcome; a planner compares possible actions; a controller selects actions within its available interface. A single agent may combine roles, but the evaluation still needs to distinguish what each contribution does.

Specify the information available to the agent and compare its performance with an appropriate simpler baseline. An agent that exploits the simulation's reward or has hidden task information has not thereby solved the underlying biological problem.

A planner needs a declared set of candidate actions and a prediction of their consequences; a controller also needs an output route. Compare each contribution against no agent and a simpler rule under the same information and evaluation conditions.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

5.3: Neural Composite

Slide 72 | Reveals 86

The Neural Composite brings declared neural observations and their context into a common representation. Which measurements entered it, how were their clocks aligned, and which signals were missing? A composite cannot turn unrecorded activity into a measured observation.

Keep each modality and its uncertainty identifiable. Test what the combined information predicts on reserved data, including comparisons that remove or simplify an input. The pictured composite remains a proposed component, not a validated model of a disease.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

5.4: Physics Engine

Slide 73 | Reveals 87

The Physics Engine describes a body's proposed mechanical behaviour under declared constraints. A movement depends on geometry, joint limits, contact, loads and timing. A plausible trajectory can still arise from the wrong forces or an unsupported controller.

Compare the simulated outcome with independent measurements in the same task and units. Keep prescribed movement separate from movement produced by a tested control model. The engine shown here is a proposed component, not proof of biological or clinical function.

Reveal Cues

1. Show the numbered step and illustrative 3D view together. Explain the proposed experiment and its limits.

5.5: World Model

Slide 74 | Reveals 88

The World Model supplies task context: where the body is, what it can encounter and how actions change the situation. It must receive only information that would be available in the declared experiment. Otherwise the agent can succeed by using a hidden advantage rather than solving the intended task.

Link the neural observations, mechanics and world to the same clock and independently scored outcome. Reserve conditions for testing and report failures as well as successes. A result inside this simulated world remains a simulation result until biological evidence supports a further claim.

This AI-generated illustration depicts parallel movement possibilities over time; it is not a measured simulation or a biological result.

A world model sets the context in which candidate actions are evaluated. Its predictions should be compared with independently measured task outcomes under conditions withheld from fitting; a still image of parallel poses does not demonstrate a changing simulation or an intervention.

Reveal Cues

1. Show the numbered step and AI-generated movement illustration together. Distinguish the illustration from measured simulation and biological results.

References

AI-generated illustration created for this presentation; not measured simulation data.

Muscle Function First

Slide 75 | Reveals 89

A hand gives us something to check. It moves through space, touches an object and sometimes misses. Muscle function is therefore a practical starting point for connecting electrical observations to mechanics and to a task outcome. Even here, the variables refuse to collapse into one another: EMG is an electrical measurement, force is a mechanical quantity, and a joint angle is a description of position.

The inside view lets us ask how a proposed muscle action could produce the movement we observe. A muscle's pull depends on its attachment, the joint's configuration, the load and the other tissues acting with it. Several combinations can produce similar visible motion. A model that fits the hand's path therefore has not uniquely identified every muscle contribution, much less the complete neural command. Independent measurements and comparisons have to narrow the alternatives.

Organ Function Next widens the ambition. A heart, gut or bladder adds different dynamics, local regulation and meaningful endpoints. Reusing a learning framework might be sensible; reusing an unsupported assumption would not be. Each organ would need its own observation model, units, time scale and independent measure of function. A movement fit cannot stand in for those tests.

The order matters because a bounded success gives us a tool and a known domain. It does not make the rest of the body a solved problem. The next organ should be chosen for an answerable question and a measurement that can expose the model's error, rather than because the diagram leaves a convenient space to colour in.

Reveal Cues

1. Show the title, image and subtitle together. Use the inside view to explain mechanics, then introduce organ function as a separately measured extension.

References

Farina, Merletti and Enoka (2014), The extraction of neural strategies from the surface EMG: an update https://pubmed.ncbi.nlm.nih.gov/25277737/

NICHD, What are the parts of the nervous system? https://www.nichd.nih.gov/health/topics/neuro/conditioninfo/Pages/parts.aspx

Myoception

Slide 76 | Reveals 90

Myoception is the working label here for sensing related to muscle. During a reach, information associated with muscles can help constrain what the limb is doing, but a muscle electrical recording is not itself a direct recording of every sensory pathway or every force.

Choose the muscle-related quantity, measurement route and task before interpreting a signal. A model can predict a declared observation and still miss an unmeasured sensory process. Keep the physical recording, inferred state and any proposed perception distinct; this slide names an inquiry rather than a validated complete map.

Reveal Cues

1. Show Myoception and its illustration. Define the working scope and distinguish muscle activity from sensory measurement.

References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

Visceroception

Slide 77 | Reveals 91

Visceroception is the working label here for sensing related to internal organs. Heartbeat, breathing and gut activity are familiar examples of bodily processes that can contribute to information about internal state. The organ's activity, a neural signal associated with it and a person's report are different observations.

To study one relationship, specify the organ, the signal, the state and the outcome. A synchronized observation can support a bounded prediction; it does not by itself reveal an entire internal experience or establish a complete organ-to-brain pathway. This working focus can overlap with muscle-related sensing, including in organs that contain muscle.

The front and back views show selected internal structures, including the stomach and kidneys. They are illustrative and incomplete: the fit and positions of structures combined from different anatomical atlases have not been validated. The images are not a complete anatomical map or a measurement of organ-to-brain signalling.

Reveal Cues

1. Show Visceroception and its illustration. Give a concrete organ example and distinguish its activity, neural measurements and subjective report.

References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

Then We Do It In Humans

Slide 78 | Reveals 92

Then We Do It In Humans is a proposed next step, not a result carried across the species boundary. A human nervous system has its own anatomy, development, disease histories and behaviour. A relationship learned in an animal may suggest a human measurement or mechanism; it cannot settle how accurately the relationship will hold in a person.

Whole-Body Emulation makes the scale of the ambition visible. To make that phrase useful, declare the first piece of the body, the first action and the first observations. A model could be accurate for a hand task while missing fatigue, balance or sensation. Those omissions define what the result means. A whole-body mesh, even a very convincing one, establishes neither whole-body measurement nor a functional emulator of everything inside it.

Human translation would begin with the measurements actually available and the independent outcome they can predict. Non-invasive recordings and movement observations could support a bounded study without assuming that an earlier invasive animal interface has a human equivalent. A later physical interface would need separate qualification; the proposed sequence supplies no procedure or permission for it.

There are precedents for carefully specified human bridges. Lorach and colleagues reported an implanted brain–spine interface enabling standing and walking in one participant with chronic tetraplegia. That is a substantial, particular result. It cannot be stretched into proof that our platform works, that every neurological condition responds, or that a whole human nervous system has been emulated. The useful inheritance is a sharper question about which connection we would test next.

Reveal Cues

1. Show the title, image and subtitle together. Explain the whole-body ambition while keeping the first bounded human task and missing coverage explicit.

References

Lorach et al. (2023), Walking naturally after spinal cord injury using a brain-spine interface
https://pubmed.ncbi.nlm.nih.gov/37225984/

Disorder

Slide 79 | Reveals 93-94

Return to the condition with a better instrument. Earlier, Disorder to Disease named the move from an observed difficulty toward an explanation. After the loop, the same arrow has a more demanding meaning: can the added measurements distinguish a disease-relevant relationship that the simpler description missed?

An average score might hide two people who fail for different reasons. One may have a timing problem; another may have a mechanical restriction. A proposed model could separate those possibilities if it predicts independently measured differences and survives the relevant controls. Assigning people to colourful clusters is not enough. The clusters must answer a question that the measurements can support.

The evaluation should separate recognition, mechanism and intervention. Recognising a recorded condition is a classification result. Predicting its behaviour under a reserved change is a stronger test of the proposed model. Changing a biological mechanism and improving a meaningful outcome would require further causal and clinical evidence. None of these stages arrives simply because the architecture contains a world model.

The repeated arrow is therefore a return to the same problem, with clearer grounds for a possible answer. What exactly changed when the difficulty appeared? What did the model predict before seeing the outcome? Which competing explanation still fits? Those questions can make the next experiment narrower and more informative. They do not require us to announce a cure while we are still deciding what the instrument measures.

Reveal Cues

1. Reintroduce Disorder and its attached arrow as the original observable difficulty.
2. Add Disease; ask what independent new prediction supports a condition-specific explanation.

Maximally Invasive

Slide 80 | Reveals 95-96

The proposed progression moves from maximally invasive toward minimally intrusive approaches. Those words describe an ambition to reduce the burden of measurement or intervention; they do not by themselves specify a device, procedure or risk level.

Invasiveness and intrusiveness are related but different. A surgical implant, a surface electrode and a camera have different physical routes and different effects on comfort, privacy, behaviour and daily use. Each proposed method needs its own evidence, appropriate oversight and comparison. A sensor outside the body is not automatically accurate, harmless or acceptable for every person.

A less intrusive system must still measure the information needed for the declared task. Compare performance, uncertainty and burden under appropriate conditions rather than assuming that removing a component preserves its function. The arrow is a research direction, not proof that the programme has already achieved either endpoint.

Reveal Cues

1. Reveal Maximally Invasive with its outgoing down arrow; define the proposed starting contrast without prescribing a procedure.

2. Add Minimally Intrusive; distinguish burden from accuracy and require an appropriate comparison.

References

Brain2Body programme ethos, current source

Rehabilitation

Slide 81 | Reveals 97-98

Rehabilitation asks how practice and support can improve a person's function. Restoration raises the outcome we want: recovering a useful ability. The arrow names that ambition, while the experiment has to tell us what changed, how long it lasted and what support was still present.

A hand that opens with external help can be valuable even before the person can open it independently. We should record that outcome accurately. A gain that remains after trainer-specific guidance ends is another result; a gain that carries into an unused task or daily activity is another again. Removing guidance should not mean removing an essential compensatory bridge, because that would change more than the learning question.

Biasiucci and colleagues tested a particular clinical route in chronic stroke. Decoded attempts at hand extension triggered functional electrical stimulation: electrical activation of peripheral motor pathways produced the corresponding muscle action. The sham comparison lacked that brain-linked contingency. This tests whether relating the attempted action to the delivered response matters, rather than merely showing a stimulation effect. The authors reported motor gains retained at follow-up; those gains do not by themselves establish a complete plasticity mechanism or qualify our proposed equipment across neurological conditions.

For this programme, restoration would need a declared baseline, matched exposure, an independently assessed functional endpoint and a suitable follow-up. Participants' own priorities would matter alongside a laboratory score. A faster virtual reach may contribute to a future study; it cannot stand in for a recovered everyday action. The strongest promise we can make now is to design a test in which assistance, retained improvement and clinical benefit cannot quietly exchange names.

Reveal Cues

1. Show Rehabilitation with its outgoing down arrow.

2. Add Restoration; separate support during action, retained function and clinical benefit.

References

Biasiucci et al. (2018), Brain-actuated functional electrical stimulation elicits lasting arm motor recovery after stroke https://pubmed.ncbi.nlm.nih.gov/29925890/

Enablement

Slide 82 | Reveals 99-100

Enablement can give someone a way to do a task they could not otherwise complete. Enhancement asks whether a defined ability can improve beyond the chosen comparison. Both require us to say who is doing what, under which conditions, and with how much help.

A body-controlled game may enable interaction without changing the person's underlying skill. An agent may improve a score by suggesting the next move. A model might reduce the time needed to reach a stated criterion. Those are different outcomes, and each can be worth pursuing. The error begins when we turn an assisted score into an unassisted ability, or one improved metric into an enhancement of the whole person.

To test faster learning, keep the end point and practice exposure comparable. Measure accuracy and timing together where both matter; a rapid mistake is not the same achievement as a rapid correct action. Stop trainer-specific information for the retention comparison, preserve a necessary bridge, and include a reserved task variation. That design would let us ask whether the benefit is immediate assistance, durable task learning or something that transfers.

The programme's skill examples make this distinction concrete. A successful virtual punch is a game outcome. A reproducible control signal is an interface outcome. A biological skill retained after guidance is a learning outcome. The progression is ambitious enough without letting one result impersonate all three. Enhancement remains a proposed question until the chosen comparison and independent evidence can support it.

Reveal Cues

1. Introduce Enablement and the arrow to the next proposed ambition.
2. Add Enhancement; define fair comparison, retained performance and task-transfer limits.

ElectroMagnetism Became Everything

Slide 83 | Reveals 101

Faraday came from modest circumstances, not from a vacuum. He was the son of a blacksmith, trained as a bookbinder and found his way into the Royal Institution's laboratory. Books, lectures, other people and access to apparatus all mattered. The compressed subtitle carries an underdog story; the history is more interesting when we leave its actual supports intact.

His experiments on electromagnetic induction in 1831 connected a changing magnetic condition with an electrical effect. A relationship that could be produced, observed and investigated became part of the foundations on which later electrical technology was built. Faraday did not single-handedly invent every device that followed, and electromagnetism is not literally everything. The title points to its extraordinary reach through everyday technology.

The relevant example is what happened to an uncertain relationship when someone found a way to interrogate it. That is the standard worth borrowing for the nervous system. Give an invisible process an observable consequence. Preserve the conditions that produced it. Find out whether it repeats, and which change makes it fail.

An ambitious biography cannot predict the success of our programme. It can make the demand clearer. If we want a useful brain–body instrument, the proposal must eventually become an apparatus, an acquisition, a comparison and a result that survives someone else's inspection. Help and funding matter because those steps take people and tools. The history earns the invitation when it leads us back to the experiment.

Reveal Cues

1. Show the title, image and subtitle together. Connect induction and later technology to the need for a checkable brain–body instrument, without promising the same historical outcome.

References

Royal Institution, Michael Faraday (1791–1867) https://www.rigb.org/explore-science/explore/person/michael-faraday-1791-1867

2 Papers

Slide 84 | Reveals 102

Two papers would give the argument a form that can be read slowly and disputed precisely. A talk can connect the programme's ideas through one action and a sequence of images. A paper has to show where each claim comes from, what the proposed method would measure, and which result would count against the explanation.

The two titles divide related jobs. NeuroTranslation concerns the mapping between specified neural and bodily observations. Brain2Body concerns the proposed interface through which declared information and feedback could move. Their connection matters: a mapping can predict without delivering an output, while an interface can exchange signals without establishing the correct interpretation of them.

These papers frame two linked questions: how measurements can constrain a mapping, and what a physical interface would need to use it. Their proposals require independent evidence before they can establish an experiment, an effective interface or a clinical result.

The useful invitation is to inspect the questions already on the page. Look for the units, source and target, independent outcome, held-out comparison and uncertainty. Look for the gap between a simulation and a biological result. A paper that makes those gaps visible is more useful than a cover that merely looks finished. When future work is authorized, the same argument must survive complete methods, references and negative results; the title cannot perform that work on its own.

Reveal Cues

1. Show the title, image and subtitle together. Introduce the two static paper previews without treating the overview image as either manuscript or claiming completed biological results.

NeuroTranslation: A New Paradigm In Brain-Body Mapping

Slide 85 | Reveals 103

A new paradigm has to change a question we can answer. NeuroTranslation proposes that neural and bodily observations should constrain one another within a declared task. Record what the instruments actually measure, predict an independent outcome, then test whether the relationship survives beyond the examples used to build it.

For one reach, the mapping could connect a permitted window of electrical observations with a later movement. The data need a common clock, clear units and a coordinate frame. The independent reference tells us where the prediction fails. A model that reproduces a training sequence has answered a narrower question than one that predicts an unused session or task variation.

The reverse direction remains informative without becoming unique. Several neural or muscle patterns can be consistent with a similar movement. A useful mapping may describe that ambiguity and show what an extra measurement resolves. It cannot promise a complete hidden state because its reconstruction looks plausible.

The paper proposes a change in approach. It earns that claim when independent observations support the relationship, with failures and uncertainty reported alongside any successes.

Reveal Cues

1. Show the paper title and image together. State the mapping question and its evidential limits.

Brain2Body: The Next Generation In Body-Brain Interfacing

Slide 86 | Reveals 104

A mapping tells us how observations relate. An interface supplies an actual route for information. Brain2Body asks what a declared body–brain connection would need before the relationship can be used rather than merely drawn.

The first requirement is to keep reading separate from writing. An EEG or EMG instrument records. A camera records movement. A software model can use those observations to drive a virtual body or provide ordinary feedback, but a biological output needs an identified and separately qualified interface. Predicting the signal that might produce an effect is not the same act as delivering it and measuring the effect.

The second requirement is to keep assistance separate from learning. If an interface helps someone complete a movement, report the assistance and its conditions. To claim retained learning, stop the trainer's information, preserve any necessary bridge and test later performance with an unused variation. That distinction determines what the interface accomplished and which experiment should follow.

Next Generation is the title’s ambition. A functioning whole-body interface, a safe intervention and improved clinical function each need their own evidence. Existing bridge research shows that specific routes can be tested; this proposed route would still have to earn its own result.

Reveal Cues

1. Read the exact paper title; explain the proposed interface question and separate it from achieved function.

References

Lorach et al. (2023), Walking naturally after spinal cord injury using a brain-spine interface
https://pubmed.ncbi.nlm.nih.gov/37225984/

Starting A New Field

Slide 87 | Reveals 105

A field starts with a question that existing tools do not settle. Here the question is how information about the body's internal condition relates to the neural pathways that carry and process it. The proposed name gives that work a home; it does not make the answer true.

The work needs measurements, maps and tests that can disagree. Which signal was recorded, where was the connection observed, and does the proposed relationship predict something new? Keep the organism, task, timing, uncertainty and comparison with simpler explanations visible. A name becomes useful when the questions behind it can be checked.

The optical laboratory microscope image is an analogy for measurement opening a question; it is not evidence that the proposed experiments have been performed.

Reveal Cues

1. Show the title, image and subtitle together. Introduce the proposal as a research field, not a completed map or established external standard.

References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

Kitmondo LAB, laboratory microscope image, CC BY 2.0; http://www.kitmondo.com/lab-medical-bioscience-equipment https://commons.wikimedia.org/wiki/File:Nikon_Optiphot_Phase_Contrast_Trinocular_Laboratory_Microscope_(15983505825).jpg

Interoceptomics

Slide 88 | Reveals 106

Interoceptomics combines two questions. Interoception concerns information about the body's internal condition; connectomics concerns mapping neural connections. Put them together and we can ask how a particular internal-body signal relates to the pathways involved in carrying or processing it.

A connection constrains an explanation but does not establish its function. A heart-related measurement, a mapped pathway and a reported sensation are not interchangeable. A useful study must declare which observations it links and test whether that link predicts an independently measured outcome. Interoceptomics is our proposed field, not a completed or universally accepted map.

The body cutaway illustrates selected muscle and organ structures, with vivid pink colouring on the exterior and a paler head shown for context. It is not a complete map of sensory pathways or evidence that their function has been measured; the head tint is not a facial-muscle map.

Reveal Cues

1. Show the title, image and subtitle together. Define both component terms and explain that the proposed field requires measured functional tests.

References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

With Your Help

Slide 89 | Reveals 107

Help becomes meaningful when it makes an answer possible. The immediate aim would be a bounded experiment that connects declared observations to an independently measured action, rather than a collection of impressive instruments waiting for a question.

That means choosing the task, specifying the acquisition and preserving a comparison capable of exposing failure. Equipment has to be readable on a common clock; the body model must fit the organism and task; the learner must be prevented from seeing its reserved targets. Useful expertise can arrive at any of those points: acquisition, biomechanics, model design, experimental control or a rigorous objection that saves us from testing the wrong thing.

A biological output route would require its own identified interface and evidence. Recording and virtual modelling alone do not establish a qualified stimulation device, actuator or clinical intervention.

The help being invited should make the next claim more answerable. Can a simpler baseline already do the job? Does another recording add independent information? Does a proposed gain remain after guidance ends? Those questions can produce useful negative results as well as positive ones. The programme's ambition survives scrutiny by becoming specific enough to fail. That is the work a future funded study would have to support.

Reveal Cues

1. Show the title, image and subtitle together. Make the invitation concrete through experiment, comparison and independent outcome; announce no invented cost.

Kung Fu THE "EASY" WAY

Slide 90 | Reveals 108

The small vision gives the wider programme a task somebody can actually attempt. Instead of beginning with an entire nervous system, begin with a hand action, a coordination task or a digital opponent. The proposed non-invasive measurements can then be judged by whether they help predict or control that declared action.

The quotation marks around EASY matter. Putting sensors outside the body does not remove timing errors, movement artefacts or the problem of interpretation. A scalp recording mixes electrical sources; a muscle recording samples muscle activity; motion capture observes the movement under its acquisition limits. Each has to contribute more than a familiar label on a wiring diagram.

A practical first comparison would distinguish ordinary motion control from a model with an independently demonstrated neural contribution. If the camera or EMG already predicts the game action, adding EEG to the apparatus does not establish that the brain recording helped. Declare the inputs, hold out appropriate data and compare the routes on the same task and endpoint. The combined model should earn its extra complexity.

This path could yield a useful game or measurement tool before any claim of accelerated biological learning. Those are worthwhile milestones with different tests. Physical assistance remains a later, separately qualified question. The small vision makes the next error easier to locate; it does not make the ultimate kung-fu ambition an achieved download. Start with the observable difference that would tell us the added measurement is doing useful work.

Reveal Cues

1. Show the title, image and subtitle together. Explain the bounded non-invasive task and the comparison needed to show that each added measurement helps.

References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

Beat Mike Tyson

Slide 91 | Reveals 109

Mike Tyson is an unforgiving benchmark in the imagination. That is why the reference works: nobody needs a lengthy explanation of the gap between an ordinary novice and a highly skilled boxer. The proposed opponent is a digital or illustrative challenge. There is no actual bout, head-contact experiment or instruction to harm anyone. The Concussion poster is a film reference; it does not turn the proposed game into a real bout or injury study.

The game can still force a precise question. Can the chosen control route produce the required virtual action at the right time? Does a predictor anticipate a declared event before the outcome is available? Does an agent improve performance by supplying information the player can use? A score only answers the question supported by the route and comparison behind it.

If the system reads the person's arm movement and moves an avatar, report body-controlled play. If it uses EEG, test whether the recording contributes independently after accounting for movement-related information and artefacts. If an agent performs part of the task, report what the agent controls. Beating a simulation under one of those conditions would not mean acquiring Mike Tyson's biological skill.

The subtitle is the joke's exaggeration. The experiment belongs in a declared virtual environment where timing, action accuracy and errors can be measured without making injury the endpoint. A future learning claim would need matched practice, trainer-off retention and transfer to a reserved variation. The challenge is useful when its familiarity makes those comparisons clearer, rather than allowing the spectacle to replace them.

Reveal Cues

1. Show the title, film poster and subtitle together. Let the subtitle land, then state the virtual task, contributor comparison and non-harm scope.

References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

1. Map Brain2Body

Slide 92 | Reveals 110-114

Map Brain2Body starts with the relationship between specified recordings and an independently measured action. It asks what the data predict, on which task, with which inputs.

Map Body2Body then asks how a declared action or relationship carries between bodies. Different proportions and mechanics make that a mapping problem of its own; one copied trajectory need not mean the same movement in another body.

Fight & Survive Mike Tyson brings the mapping into the proposed digital challenge. A control route must work within the game's timing and action demands. The point is to expose a failure that a smooth prerecorded animation could conceal. Survival here is a game outcome, never a biological harm endpoint.

Beat & Become Mike Tyson deliberately raises the ambition from getting through a task to learning from a skilled reference. Become cannot mean copied identity or the transfer of a complete career's expertise. It asks whether useful task information would improve a trainee's performance and whether any improvement remains when trainer-specific help stops. Preserve any necessary bridge so that the retention test changes the trainer's contribution rather than the trainee's basic access.

Write Motorsensation Is All You Need is the proposed scientific return. We would compare a model that relates action to its declared sensory consequences with a simpler model, using reserved outcomes and matched information. The mistakes matter as much as the successes: which sensory measurements are missing, and which predictions fail when the task changes? Each step has to supply evidence the next one can use.

Reveal Cues

1. Reveal item 1, Map Brain2Body; define source, target, units and held-out action.
2. Add item 2, Map Body2Body; introduce the separate cross-body mapping problem.
3. Add item 3, Fight & Survive Mike Tyson; introduce the digital challenge.
4. Add item 4, Beat & Become Mike Tyson; distinguish game success from retained learning and identity.
5. Add item 5, Write Motorsensation Is All You Need; state the comparative research question.

NeuroHandler

Slide 93 | Reveals 115

A hand action is small enough to inspect and complicated enough to expose an error. NeuroHandler proposes manual dexterity as the first task family: the earlier hand rubbing, knuckle gesture and interlaced-hand stretch give us observable actions to define. The action must include its timing and return, rather than one selected pose that happens to look correct.

The measurement roles remain separate. EEG records scalp electrical potentials. Surface EMG records electrical activity generated by muscle fibres, without becoming a direct peripheral nerve recording. Motion capture supplies an independent observation of the moving hand and body. The proposed left and right camera views must be calibrated and synchronized, with useful extra views where occlusion leaves gaps; the two views contribute to one joined motion map only after that alignment has been checked.

EMS is Electric Muscle Stimulation: an electrical output intended to activate muscle. EEG and surface EMG read signals; EMS supplies a stimulus; cameras observe the resulting movement. Those roles must stay separate. A stimulation pulse can contaminate the recordings, so a claimed response needs timing, artefact controls and a declared endpoint. This is a proposed interface, not evidence that stimulation has written a skill into the nervous system.

The first model comparison can still be stated. Predict a reserved movement or task outcome from declared inputs, compare against a simpler route, and test whether each added measurement helps. Hand contact, force, electrical activity and visible movement are different observations; an unmeasured one must stay unmeasured. A model that predicts the action is a prediction result. Improved dexterity and retained learning would require further tests.

Reveal Cues

1. Show the title, image and subtitle together. Explain Manual Dexterity with the declared hand action, synchronized left/right capture and independent outcome.

References

Farina, Merletti and Enoka (2014), The extraction of neural strategies from the surface EMG: an update https://pubmed.ncbi.nlm.nih.gov/25277737/

OpenSim documentation, How Inverse Kinematics Works https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090047/How+Inverse+Kinematics+Works

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

NeuroJuggler

Slide 94 | Reveals 116

Moving each hand well is not enough to juggle. One releases while the other prepares to catch; the object has a flight of its own. NeuroJuggler puts the relationship between limbs into the outcome, which makes it a useful next question after manual dexterity.

The proposed progression begins with a transfer between hands and may extend to two-ball juggling. The model must account for release, flight, catch and the next action on a common clock. A hand trajectory can look good in isolation while the object arrives outside the useful timing window. The comparison therefore needs independent object observations as well as the declared limb measurements.

The inherited EEG, surface EMG, EMS and motion-capture roles remain distinct. Left and right camera observations need a shared calibration, pose identity and clock, with extra views for the object and occluded hands where needed. EMS is the proposed muscle-output channel. Recording and stimulation artefacts must be tested against a simpler explanation for apparent prediction or coordination.

A useful model would predict the joint task better than a suitable simpler baseline on reserved trials or conditions. That does not require us to claim that it has reconstructed the complete nervous system. The specific test is what the added observations contribute to coordination, after preserving the information limits. Improved play with guidance, retained coordination after guidance and generalisation to a changed pattern would each need their own outcome. A dropped ball can be more informative than a flawless animation.

Reveal Cues

1. Show the title, image and subtitle together. Explain Multi-Limb Coordination; connect hand-to-hand transfer, object timing and independently measured joint outcomes.

References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

NeuroBoxer

Slide 95 | Reveals 117

NeuroBoxer makes timing hard to hide. The proposed digital opponent creates events to which an avatar can respond, while the measurement route determines what we are entitled to say about the response. A movement-controlled game, a neural prediction and an assisted biological action are separate experiments.

The illustration shows the participant with arms raised in a boxing pose. It is a proposed measurement setup, not evidence of a working interface or copied skill.

Reveal Cues

1. Show the title, image and subtitle together. Beat Mike Tyson names the virtual challenge, not a real bout.

References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

Farina, Merletti and Enoka (2014), The extraction of neural strategies from the surface EMG: an update
https://pubmed.ncbi.nlm.nih.gov/25277737/

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

NeuroBoxer: Read

Slide 96 | Reveals 118

Read means acquire declared observations. EEG can contribute scalp potentials; surface EMG can contribute muscle electrical activity; motion capture can contribute limb and body movement. All require time alignment and their own acquisition errors. A successful EEG-labelled predictor could still exploit movement-related contamination or another unintended cue, so any claim of an independent neural contribution needs the appropriate restricted-input comparisons and controls.

The illustrated participant raises the arms in a boxing pose; calibrated movement capture helps relate the recorded movement to the declared signals.

Reveal Cues

1. Read: identify EEG, muscle EMG and calibrated motion observations with a shared clock.

References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

Farina, Merletti and Enoka (2014), The extraction of neural strategies from the surface EMG: an update
https://pubmed.ncbi.nlm.nih.gov/25277737/

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

NeuroBoxer: Write

Slide 97 | Reveals 119

Write must name its destination. Software can drive an avatar or a display. EMS—Electric Muscle Stimulation—is the proposed route for stimulating muscle in a physical version. A pulse and a movement do not establish a copied motor skill, a selected internal sensation or a direct synaptic instruction. The response, timing and stimulation artefact need separate measurement; the illustration supplies no treatment protocol.

The Write illustration keeps the participant's arms by the sides, omits EMG and EMS sleeves, and retains the EEG cap and two side cameras. The television shows first-person digital boxing-gloved arms. Any physical output route would require separate qualification; the illustration does not establish one.

Reveal Cues

1. Write: distinguish software/avatar output from an unqualified biological route; EMS remains a proposed muscle-output route requiring separate validation.

References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

Farina, Merletti and Enoka (2014), The extraction of neural strategies from the surface EMG: an update
https://pubmed.ncbi.nlm.nih.gov/25277737/

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

NeuroBoxer: React

Slide 98 | Reveals 120

React closes the useful task comparison. Measure the avatar's response time and action result against independently logged game events. State whether the person, agent or a combination chose the action. If the eventual question is learning, compare equivalent practice and test performance after the trainer's information ends, with any essential bridge unchanged. Mike Tyson remains the virtual reference. Game success would neither expose anyone to a real opponent nor prove the transfer of his boxing skill.

The React illustration shows raised physical arms, body sleeves and cameras without an EEG cap. It depicts proposed body-directed output, which requires a separately qualified route and evidence; the illustration does not demonstrate an effective intervention.

Reveal Cues

1. React: measure response timing and outcome, then distinguish assistance from trainer-off retention.

References

MNE documentation, Repairing artifacts with ICA
https://mne.tools/stable/auto_tutorials/preprocessing/40_artifact_correction_ica.html

Farina, Merletti and Enoka (2014), The extraction of neural strategies from the surface EMG: an update
https://pubmed.ncbi.nlm.nih.gov/25277737/

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

NeuroArcher

Slide 99 | Reveals 121

An archer's two hands do different jobs in one action. One establishes the bow's orientation; the other draws and releases. NeuroArcher asks whether a proposed model can preserve that coordination and predict the outcome in a declared virtual task. Shoot Robin Hood is the precision joke: an illustrative target challenge, not a person or animal to shoot.

The illustration shows a participant with a bow and arrow. It is a proposed measurement setup, not evidence of a working interface or transferred skill.

Reveal Cues

1. Show the title, image and subtitle together. Shoot Robin Hood is an illustrative precision challenge, with no actual person or animal targeted.

References

OpenSim documentation, How Inverse Kinematics Works
https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090047/How+Inverse+Kinematics+Works

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

NeuroArcher: Read

Slide 100 | Reveals 122

Read should capture the variables needed to test that relationship. The inherited EEG and EMG remain recordings with distinct meanings. Motion capture needs both limbs, the relevant object state and the release event on the same clock. Left and right camera views can reduce missing geometry only after they are calibrated into one frame; two disagreeing views do not cancel each other's error. If force or contact is not measured, the animation cannot manufacture a measured value for it.

The illustration shows the archer raising the bow and arrow while wearing the EEG cap. Motion capture can help align the physical movement with the declared observations.

Reveal Cues

1. Read: align both limbs, object state and release with the declared observations.

References

OpenSim documentation, How Inverse Kinematics Works
https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090047/How+Inverse+Kinematics+Works

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

NeuroArcher: Write

Slide 101 | Reveals 123

Write again needs an endpoint. A command to a virtual archer is a software action; visual feedback is an ordinary sensory route. EMS adds the proposed muscle-stimulation channel. None establishes direct delivery of an intention or an instant retained skill. Output timing, artefacts and task performance need their own checks. The bow and target are illustrative equipment; the scene supplies no physical training or stimulation procedure.

The Write illustration keeps the participant’s arms by the sides, omits EMG and EMS sleeves, and retains the EEG cap and two side cameras. The television shows first-person digital arms holding a bow. Any physical output route would require separate qualification; the illustration does not establish one.

Reveal Cues

1. Write: name virtual output or ordinary feedback; do not equate EMS with a validated biological writer.

References

OpenSim documentation, How Inverse Kinematics Works
https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090047/How+Inverse+Kinematics+Works

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

NeuroArcher: React

Slide 102 | Reveals 124

React tests the relationship among the game cue, the selected action, release timing and target error. An agent may improve aim by doing the aiming; the result must name that contribution. A retained-learning claim would instead require a later comparison without trainer-specific guidance and a reserved variation. The target makes precision visible. The declared units, comparison and errors tell us whether precision improved.

The React illustration shows raised physical arms, body sleeves and cameras without an EEG cap. It depicts proposed body-directed output, which requires a separately qualified route and evidence; the illustration does not demonstrate an effective intervention.

Reveal Cues

1. React: measure release timing and target error; separate agent assistance and retained human learning.

References

OpenSim documentation, How Inverse Kinematics Works
https://opensimconfluence.atlassian.net/wiki/spaces/OpenSim/pages/53090047/How+Inverse+Kinematics+Works

Osuagwu, Whicher and Shirley (2020), Active proportional electromyogram controlled functional electrical stimulation system https://pmc.ncbi.nlm.nih.gov/articles/PMC7718906/

Coming Eventually

Slide 103 | Reveals 125

An action changes what the body senses. Fingers contact an object, a limb changes position, and an expected event either happens or fails to happen. Motorsensation Is All You Need asks whether modelling declared actions with their sensory consequences improves a specified prediction or control task.

The phrase All You Need is a provocation to test. It cannot remove vision, attention, cognition, task context or internal senses from the biology. The experiment should name which sensory observations are available and which are absent. A prediction of sensation is also different from the recipient reporting or discriminating an actual delivered sensation.

A useful comparison would hold the task, outcome and information budget clear. Does the proposed action–sensory model outperform a simpler movement-only model on reserved conditions? Does the benefit survive changes of session, object or task within the stated validation domain? If the sensory target leaks into the input, apparent success cannot answer the proposed question. If an added modality helps only in one setting, that limit belongs with the result.

Coming Eventually places the paper after the proposed skill tasks. The title does not report a written manuscript, completed training or a general theory established by this programme. The contribution would be an inspectable relationship, a fair comparison and its outcome, including a negative one. A result showing where the added sensory information fails would still tell us which part of the loop the next experiment needs to measure.

The image is a screenshot of the 2017 Attention Is All You Need paper. It is a historical reference for Transformers, not the proposed Motorsensation paper.

Reveal Cues

1. Show the title, image and subtitle together. Identify the Attention Is All You Need screenshot as a historical reference, then state the proposed action/sensory comparison.

References

Vaswani et al. (2017), Attention Is All You Need https://arxiv.org/abs/1706.03762

System 1: FAST

Slide 104 | Reveals 126-128

Fast action, slower checking, then a proposed compression of what the checking discovered. That is the computational analogy behind this sequence. The familiar System 1 and System 2 language draws on Kahneman's distinction between intuitive and deliberative thought. It does not identify three measured anatomical systems inside the brain.

For a proposed machine implementation, FAST could mean running a frozen model to make a task prediction under a declared time limit. SLOW could mean a separate process that reviews errors, considers more alternatives or develops a candidate update. Keeping the two roles distinct allows the action route to remain stable while changes are tested. Neither the labels nor a smooth demonstration establish actual real-time performance.

COMPRESSED proposes retaining useful work from the slower process in a form that is cheaper or quicker to use later. A smaller model, a cached relation or another declared representation might be candidates. Each would need a matched comparison of useful latency, cost and quality on the original tests. An update that improves a training score while damaging held-out behaviour has failed the proposed improvement, however small its file becomes.

System 3 also carries the human–machine partnership framing: a person can ask the question, inspect the failures and decide what a model's result means. That is a proposed organisation of work. It is not a demonstrated neurobiological architecture, a measured compression of biological learning, or proof that this programme has made a trained model faster. The analogy earns its place by making the separation of acting, checking and accepted updating legible.

Reveal Cues

1. Reveal System 1: FAST with the down arrow; introduce the frozen-inference analogy.

2. Add System 2: SLOW with its arrow; explain independent checking and candidate updates.

3. Add System 3: COMPRESSED; explain a proposed accepted compression/partnership, without a biological or performance claim.

References

Daniel Kahneman (2011), Thinking, Fast and Slow, Microsoft Research lecture
https://www.microsoft.com/en-us/research/video/thinking-fast-and-slow/

Most Valuable Human Skill

Slide 105 | Reveals 129

The valuable skill in this argument is learning: being able to meet a new task, discover the useful relationships and retain enough of them to act again when the help disappears. The strongest version of the title is a position about the future value of that ability. It is not an empirical ranking of every human skill, and Only is the deliberate exaggeration carried by the image.

The walking-evolution silhouette is symbolic; human evolution did not proceed as a single straight line. The scientific question here is about learning. Could a particular form of feedback improve the rate at which someone reaches a defined, retained level of performance? We would need comparable practice, a meaningful endpoint and tests after trainer-specific guidance stops. Otherwise a system that simply takes over the difficult part can appear to have taught it.

Nor does a movement endpoint contain the whole skill. A musician's hand sequence, an archer's release and a boxer's response are observable parts of larger activities that involve perception, judgement and context. Improving one part can matter. The claim should remain the size of the measured part until broader transfer is demonstrated.

The invitation is to investigate learning with the same seriousness we give the hardware. Make useful help possible, then find out what the learner keeps. If the benefit disappears, we have learned that it was assistance. If it remains under a reserved change, we have a stronger reason to ask the next question. The ambition is bold enough when the experiment can distinguish those endings.

Reveal Cues

1. Show the title, image and subtitle together. Explain that the walking-evolution image is symbolic, then distinguish assistance from measured retention.

References

Smithsonian Human Origins Program, Frequently Asked Questions https://humanorigins.si.edu/education/frequently-asked-questions

Months & Years

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Months and years; weeks and days. The contrast is the ambition to make useful learning faster. It is not a measured speedup and it does not tell us which training task, person or outcome has changed.

To compare times, define the starting point and a meaningful level of performance, then measure practice exposure and the result after trainer-specific help ends. Use matched comparisons and a reserved task variation. Faster assisted performance can be valuable; a faster route to retained skill is a different claim and needs its own evidence.

Reveal Cues

1. Show the title, calendar-and-stopwatch image and subtitle together. Present the time contrast as an ambition and name the matched retention test needed to measure it.

References

Brain2Body programme ethos, current source

Brain2Body canonical glossary, revision 33

Eric Rothermel, monthly schedule image, Commons CC0
https://commons.wikimedia.org/wiki/File:Monthly_schedule_(Unsplash).jpg

Liilia Moroz, stopwatch image, Commons CC0 https://commons.wikimedia.org/wiki/File:Stopwatch_made_in_USSR.jpg

Whole-Body Emulation

Slide 107 | Reveals 131

Whole-Body Emulation returns us to the reason the body was in the argument from the beginning. A nervous system acts through a body in an environment. A model of an isolated brain leaves out many of the measurements and consequences that could help test its account of action.

That does not mean a body model automatically becomes a brain model. Geometry can constrain possible movement; mechanics can test a proposed force or trajectory; independent recording can test a relationship between observations. None of those steps establishes the missing neural dynamics by itself. Similar visible behaviour can remain consistent with different internal explanations.

An emulator needs a declared domain. Which organism and state does it represent? Which tasks, inputs and outputs are included? How close must the predictions be, in which units, and on which reserved conditions? A convincing picture and a successful replay cannot answer those questions. Accuracy concerns agreement with an independent reference; precision concerns agreement across repeated measurements under stated conditions. Neither follows from calling a model an emulator. Report the error, repeatability and uncertainty.

Whole-Brain Emulation remains the wider ambition beside the body. The path proposed here proceeds through smaller relationships that can be checked: brain observations, muscle observations, measured movement, declared virtual mechanics and a learner tested against an independent outcome. As those relationships survive new evidence, the validated domain could grow. The present programme does not claim complete functional coverage of a body or brain. The full-circle return matters because we can now point to the missing experiment at each connection.

Reveal Cues

1. Show the title, image and subtitle together. Reconnect whole-body and whole-brain emulation; reconnect the five measurement/modelling roles and distinguish validated domain from total functional coverage.

References

NIST Technical Note 1297, Appendix D.1 Terminology https://www.nist.gov/pml/nist-technical-note-1297/nist-tn-1297-appendix-d1-terminology

Vertical Column Central Axis

Slide 108 | Reveals 132

When you run, limb movement, balance, breathing and circulation have to work together. Those demands meet in overlapping brain circuits. The somatic and autonomic systems describe different bodily relationships; their integration is spread across the brain. The backslash is our visual shorthand for selected brain regions, not the name of an established tract.

Somatic function includes sensing skin, muscles and joints, and controlling skeletal movement. Primary somatosensory cortex processes body-related signals; its medial continuation includes the caudal paracentral lobule. Primary motor cortex and the rostral paracentral region contribute to skeletal movement. Premotor, supplementary motor, cingulate motor and posterior parietal territories contribute to preparing and organising actions. Their exact functional boundaries do not follow every visible groove. Much of the hand and face representation lies outside this medial view.

Thalamic nuclei relay and transform different sensory and motor signals. Sensory ventral-posterior nuclei differ from motor-related ventral-anterior and ventral-lateral nuclei. Basal-ganglia circuits contribute to action selection and movement scaling; cerebellar circuits contribute to timing, coordination and error-related adjustment. The cerebellum also contains nonmotor territories. A whole-thalamus or whole-cerebellum surface therefore gives structural context rather than one uniform functional label.

Autonomic output regulates internal effectors through sympathetic and parasympathetic pathways. Hypothalamic circuits link internal condition with autonomic, endocrine and motivated responses. Brainstem visceral sensory relays, including the nucleus of the solitary tract, receive incoming information; that afferent traffic must be distinguished from motor commands. Parasympathetic populations include the Edinger–Westphal preganglionic, salivatory, dorsal motor vagal and cardiac vagal populations. Sympathetic premotor circuits influence preganglionic neurons in the spinal cord. Those spinal and peripheral stages remain outside this brain-only image.

Integration becomes concrete where action and internal demand meet. Insular, anterior/midcingulate, medial prefrontal, amygdalar, hypothalamic, periaqueductal and parabrachial circuits participate in different aspects of body sensing, state and coordinated responses. Their contributions vary by circuit, task and organ. Primate multisynaptic tracing links selected motor and somatosensory cortical regions with the adrenal medulla. Human imaging identifies interleaved motor and action-control territories; the broader allostatic role proposed for the somato-cognitive action network remains an interpretation of those findings, rather than a complete human autonomic control map.

Breathing makes the overlap obvious. The diaphragm is skeletal muscle, yet breathing can continue automatically through brainstem respiratory networks and spinal motor pathways. Deliberate control, visceral feedback and cardiopulmonary adjustments interact with that rhythm. In mice, identified pontine circuits coordinate bladder emptying with relaxation of the striated outlet sphincter; human mechanisms require their own evidence. These are specific mechanisms to study, rather than evidence for a single universal integration centre.

The shades select somatic, autonomic/visceral and shared integration associations. The broad parcels show illustrative associations: they do not measure activity or divide the brain into exclusive compartments, and unshaded regions can also participate. The image is a constructed medial view of the source-left hemisphere, front to the right, with a 12° display roll. Twenty occupied atlas labels have no model geometry. Insula, amygdala and lateral cortex require a separately identified view or transparent treatment; tiny nuclei cannot be resolved from broad atlas parcels. The supporting outline

names the regions, subdivisions, evidence and remaining localization gaps. Most of these systems are bilateral.

Reveal Cues

1. Show the existing title, revised image and subtitle together. Identify somatic, autonomic/visceral and shared associations, then relate them through breathing; keep the single-hemisphere visibility limit explicit.

References

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Short-term training-dependent plasticity of the corticospinal diaphragm control in normal humans https://pubmed.ncbi.nlm.nih.gov/17964864/

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Blood Pressure Monitor

Slide 109 | Reveals 133

A blood-pressure monitor provides a familiar analogy: name a quantity, measure it in stated units, and make the result interpretable against an appropriate reference. The proposed nervous-system monitor would need its own declared signals, calibration, timing and independent tests. No single established number for the whole nervous system is supplied here.

Then the pacemaker: a second analogy for a separately qualified way to affect a declared physiological process. Reading a signal does not establish a safe or effective way to write one. Any future interface would need a specified target, measured response, appropriate comparisons and evidence suited to its intended use.

The proposed sequence is measurement, then a separately tested intervention. Neither analogy is a demonstrated device, treatment or skill-transfer result. A meaningful benefit would have to be measured in the task and conditions claimed.

Reveal Cues

1. Show the title, image and subtitle together. Explain the proposed measurement first and the separately qualified intervention next; claim no demonstrated device.

References

Brain2Body programme ethos, current source

That's How I Know Kung Fu

Slide 110 | Reveals 134

The opening gave us a finished download. The ending gives us the work hidden inside it. To make a skill teachable through a declared interface, we would first have to know what was observed, what was predicted, what actually reached the learner and what the learner could do afterwards.

The proposed loop now has five recognizable roles: brain signals, muscle signals, independently measured movement, a virtual body and a learning model. Keep them on the same clock and within the same task. Let the model face an outcome it did not choose. When the question becomes teaching, stop trainer-specific information, preserve a necessary bridge and test delayed performance with a reserved variation. Each step exposes a different way the attractive answer could be wrong.

That is how I know kung fu remains the film's fiction and the talk's bookend. It does not report a skill I have downloaded or promise that you will receive one. And You Will Too carries an invitation to the research ambition: help turn one more missing relationship into a question somebody can actually test.

A useful next result might be a reliable measurement, a model that predicts a declared action, a control route that works, or a failure that tells us why the next step is not ready. We should name each one accurately. There is no kung fu to download from this programme yet. There is a connected experiment to build, inspect and improve.

Reveal Cues

1. Show the title, image and subtitle together. Let the closing image land; distinguish the invitation from a guaranteed skill download and finish on the next checkable result.

eliminatenoise.org

Slide 111 | Reveals 135

Find the project at eliminatenoise.org. The experiments still have to stand on their measurements, comparisons and independently checked outcomes.

Reveal Cues

1. Show eliminatenoise.org alone, centered in regular type, as the final slide.